

Universitat Politècnica de Catalunya

Facultat de Informàtica

Complex and Social Networks

CSN Laboratory 4 : Non-Linear Regression on Dependency Trees

Davide Volpi

Domenico Azzarito

Contents

1	Introduction	2
2	Results	2
2.1	Preliminary Data Analysis	2
2.2	Heteroskedasticity Analysis	3
2.3	Model Selection and Comparison	3
2.3.1	R^2 and Residual Standard Error	3
2.3.2	Residual Sum of Squares	4
2.3.3	Akaike Information Criterion and selection based on ΔAIC	4
2.3.4	Comparison with Bayesian Information Criterion and ΔBIC	4
2.4	Best Fit Model Plots	4
3	Discussion	5
3.1	Summary of Results	5
3.2	Rejection of the Null Hypothesis	5
3.3	Heteroskedasticity and Model Validity	5
3.4	Model Selection: AIC versus BIC	5
3.5	Expected Model Behaviour	6
3.5.1	Impact of Data Distribution on Model Fit	6
3.5.2	Overall Model Assessment	6
3.6	Model Comparison: 5/5+ vs. 4+	6
3.7	Cross-Linguistic Universality and Variation	7
3.8	Conclusions	7
4	Methods	8
4.1	Data Preparation	8
4.2	The Ensemble of Models	8
4.2.1	Model Definitions	8
4.2.2	Initial Parameter Estimation	9
4.2.3	Non-linear Least Squares Fitting	9
4.3	The Heteroskedastic Problem	10
4.3.1	White's Test for Heteroskedasticity	10
4.3.2	LOWESS Smoothing for Variance Estimation	11
4.3.3	Heteroskedastic Log-Likelihood	12
4.3.4	Model Selection with Heteroskedasticity-Corrected Information Criteria	12
4.3.5	Model Performance Metrics: R^2 and RSE	13
A	Results tables	15
B	Best Fit Models Plots	24
C	Preliminary Analysis Plots	31

1 Introduction

The goal of this laboratory session was to model the relationship between sentence length and network connectivity in syntactic dependency trees across multiple languages. Specifically, we analyzed how the second moment of the degree distribution, $\langle k^2 \rangle$, scales with the number of words n in a sentence. We tested multiple candidate models, including power-law functions, exponential growth, logarithmic scaling, and mixed forms, as well as their extended versions with an additive constant term. Model selection was performed using information-theoretic criteria (AIC and BIC) that balance goodness of fit with model complexity. To account for heteroskedasticity in the variance of $\langle k^2 \rangle$ across different sentence lengths, we employed LOWESS smoothing to estimate observation-specific variances when computing the log-likelihood.

2 Results

2.1 Preliminary Data Analysis

The analysis began with an examination of the summary statistics for sentence length (n) and the second moment of the degree distribution ($\langle k^2 \rangle$) for 21 languages (Table 1). Across all languages, the sample size (N) is 1000 sentences. The mean sentence length (μ_n) varies considerably, with Finnish having one of the shortest mean lengths (15.81) and Japanese having the longest (28.79). The mean of the target metric, $\mu_{\langle k^2 \rangle}$, shows less variation, generally falling between 5 and 6, with Korean exhibiting the lowest value (4.73) and Japanese the highest (6.00).

A preliminary visualization for each language reveals important characteristics of the data. Figure 1, using Arabic as a representative example, illustrates the relationship between $\langle k^2 \rangle$ and n . The plot (bottom left) shows that nearly all data points fall within the theoretical bounds $4 - 6/n \leq \langle k^2 \rangle \leq n - 1$ (blue lines). Moreover, the empirical values consistently lie above the null-model prediction for uniformly random labelled trees (red line), given by $\langle k^2 \rangle = (1 - \frac{1}{n})(5 - \frac{6}{n})$, indicating that real dependency trees are substantially more constrained than structures generated under this random baseline.

The visualization also reveals that the dispersion of $\langle k^2 \rangle$ increases with sentence length n . This widening spread reflects the fact that larger trees admit a broader set of admissible configurations, naturally producing greater variability in $\langle k^2 \rangle$. This trend is consistent across all languages in the dataset.

Finally, the log-log plot (top right) and the averaged curves (bottom row) suggest a sub-linear, possibly power-law, relationship between $\langle k^2 \rangle$ and n , with a mild saturation effect at larger values of n . The complete set of preliminary plots for all 21 languages is provided in Appendix C.

Table 1: Summary statistics for each language. N is the number of sentences, μ_n is the mean sentence length with standard deviation σ_n , $\mu_{\langle k^2 \rangle}$ is the mean squared degree with standard deviation σ_{k^2} .

Language	N	μ_n	σ_n	$\mu_{\langle k^2 \rangle}$	σ_{k^2}
arabic	1000	20.75	8.61	5.22	0.71
chinese	1000	21.42	8.62	5.67	0.81
czech	1000	18.61	7.76	5.39	0.82
english	1000	21.18	8.21	5.81	0.83
finnish	1000	15.81	6.53	5.36	0.96
french	1000	24.73	10.01	5.76	0.68
galician	1000	23.51	9.80	5.58	0.69
german	1000	21.33	8.54	5.79	0.78
hindi	1000	23.83	9.55	5.78	0.76
icelandic	1000	18.83	7.66	5.43	0.74
indonesian	1000	19.45	7.69	5.33	0.75
italian	1000	23.73	9.85	5.71	0.70
japanese	1000	28.79	10.99	6.00	0.72
korean	1000	16.58	6.67	4.73	0.65
polish	1000	18.38	7.42	5.23	0.72
portuguese	1000	23.41	9.48	5.60	0.69
russian	1000	19.36	8.04	5.27	0.75
spanish	1000	23.28	9.44	5.54	0.63
swedish	1000	19.08	7.65	5.58	0.78
thai	1000	22.32	8.99	5.21	0.70
turkish	1000	16.88	6.63	4.97	0.71

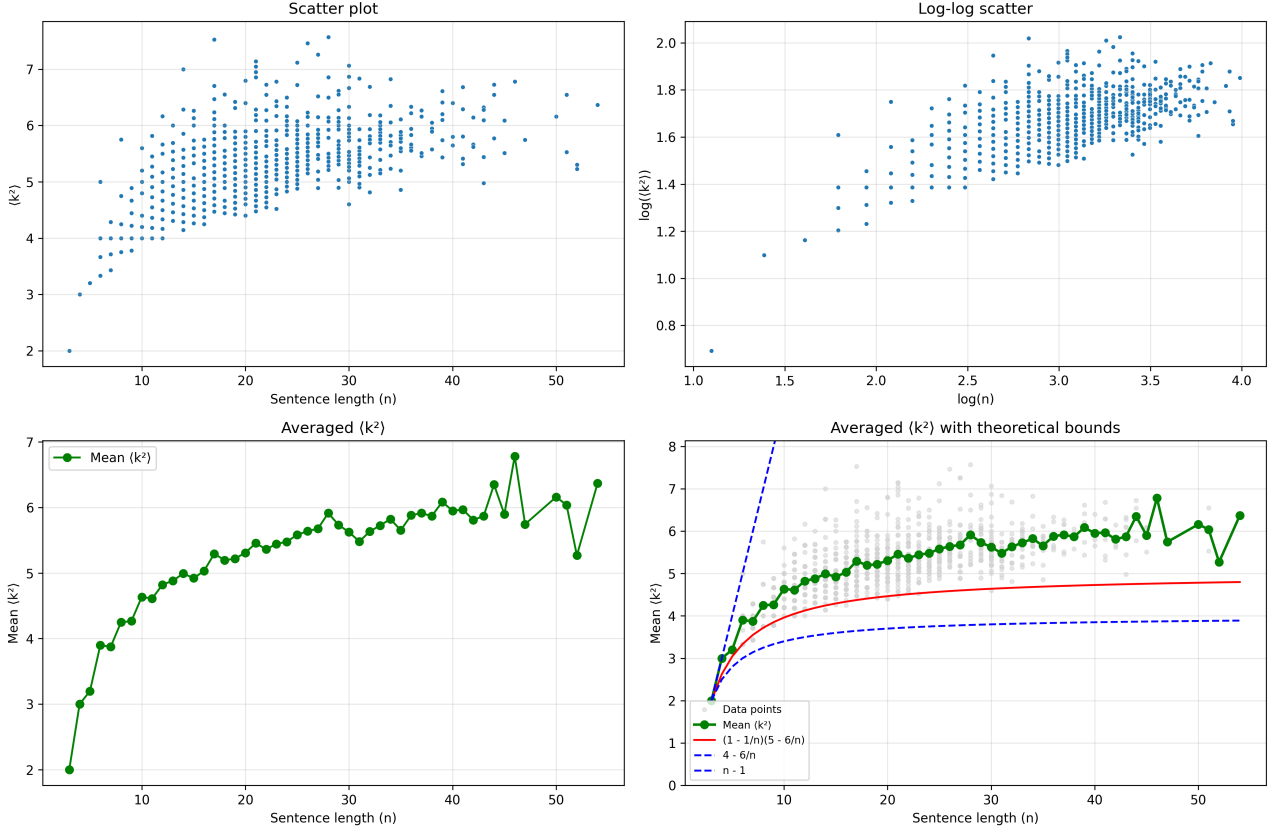
Preliminary plots for $\langle k^2 \rangle$ (arabic)

Figure 1: Preliminary results for the Arabic language language.

2.2 Heteroskedasticity Analysis

To ensure the correct interpretation and calculation of AIC, the assumption of homoskedasticity (constant error variance) was formally tested using White’s test. The results, detailed in Table 2, show the test statistic and corresponding p-value for each model across all languages.

The results are overwhelming: for the vast majority of model-language combinations, the p-values are far below the conventional significance level of 0.05 (many are $p < 10^{-10}$). This provides strong evidence to reject the null hypothesis of homoskedasticity. The data exhibits significant heteroskedasticity, where the variance of the residuals is not constant but changes with the level of the fitted values.

While a few combinations (e.g., Model 5 for English, $p=0.55$; Model 2 for Chinese, $p=0.56$) do not show statistically significant heteroskedasticity, we decided to abandon the assumption of homoskedasticity for all models to ensure a fair and consistent comparison across the entire ensemble. Therefore, a heteroskedasticity-robust method, as described in the Methods section, was used to calculate all information criteria, providing a valid basis for model selection.

2.3 Model Selection and Comparison

The full ensemble of 11 models was fitted to the data for each language using non-linear least squares regression with heteroscedasticity-corrected variance estimates (see Methods). Model performance was evaluated using the Residual Sum of Squares (RSS), Akaike Information Criterion (AIC), and Bayesian Information Criterion (BIC). To facilitate direct comparison, we calculated the differences in AIC (ΔAIC) and BIC (ΔBIC) relative to the best-performing model for each language.

2.3.1 R^2 and Residual Standard Error

Table 3 presents the coefficient of determination (R^2) for all models, while Table 4 shows the residual standard error (RSE) values. The null model consistently produces negative R^2 values (ranging from -0.88 to -3.82) and the highest RSE values (from 0.83 to 1.88), confirming its fundamental inadequacy.

Model 5 and Model 5+ achieve the highest R^2 values, typically between 0.32 and 0.55, with corresponding RSE values of 0.51 to 0.83. The addition of the offset parameter d in extended models consistently improves both metrics, though improvements are generally modest (R^2 increase of 0.01–0.02, RSE reduction of 0.01–0.02).

Cross-linguistic variation appears in both metrics: languages with higher variance in $\langle k^2 \rangle$ (e.g., Finnish: $\sigma_{k^2} = 0.96$) exhibit higher RSE values (0.83 for Model 5+), while languages with lower variance (e.g., Korean: $\sigma_{k^2} = 0.65$) show lower RSE (0.59 for Model 5+).

2.3.2 Residual Sum of Squares

The RSS values for all models across languages are presented in Table 5. Lower RSS values indicate better fit to the data. The null model consistently shows the highest RSS values, confirming its poor fit. Among alternative models, Model 5 and Model 5+ generally achieve the lowest RSS values, with Model 5+ showing marginal improvements in many cases. Notably, the RSS values for the null model are often 5–10 times larger than those of the best alternative models, demonstrating the inadequacy of the random tree hypothesis.

2.3.3 Akaike Information Criterion and selection based on ΔAIC

The AIC values for all models are shown in Table 6. AIC balances goodness of fit with model complexity, penalizing additional parameters to prevent overfitting. The null model consistently has extremely high AIC values, often orders of magnitude worse than alternative models. Model 5 and Model 5+ dominate across languages, with the choice between them depending on whether the additional parameter d in Model 5+ provides sufficient improvement to justify its inclusion.

The Table 7 presents the differences in AIC relative to the best model for each language. A ΔAIC value of 0 indicates the best model, while larger values indicate progressively worse models. It helps to compare models.

2.3.4 Comparison with Bayesian Information Criterion and ΔBIC

The Bayesian Information Criterion (BIC) provides an alternative model selection framework that can be compared directly with AIC. Table 9 presents the absolute BIC values for all models, while Table 10 shows the ΔBIC values relative to the best model for each language according to this criterion.

BIC imposes a stronger penalty on model complexity than AIC, with the penalty term proportional to $\log(N)$ rather than a constant value of 2. Given our sample size ($N = 1000$, so $\log(N) \approx 6.91 > 2$), this leads to systematically different model preferences that favor simpler functional forms.

2.4 Best Fit Model Plots

Figure 2 presents representative examples of the best-fitting model for Arabic. The complete set of best-fit plots for all 21 languages is provided in Appendix B.

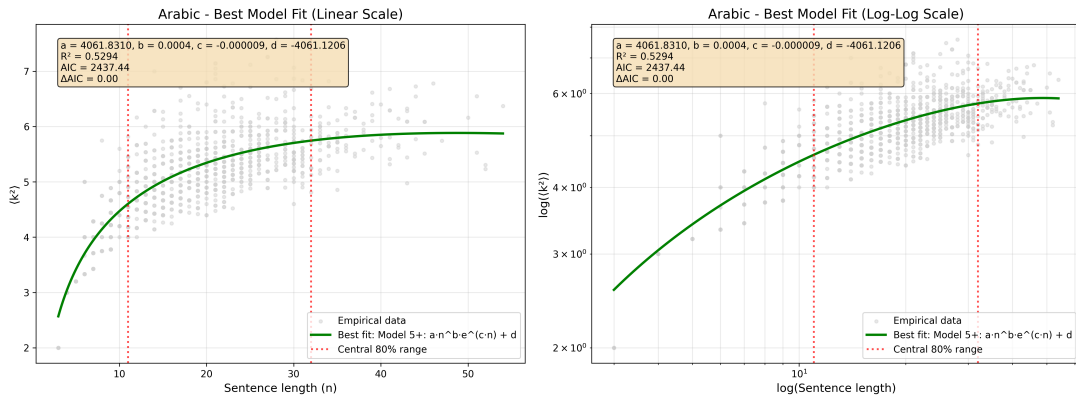


Figure 2: Arabic - Best fit model.

The quality of fit varies systematically across sentence length ranges. Within the central 80% region (bounded by red dashed lines), where approximately 800 of 1000 sentences per language are concentrated, the models achieve excellent fit quality. This region provides dense, well-distributed data that robustly constrains parameter estimation and enables accurate capture of the heteroscedastic variance structure via LOWESS smoothing.

In contrast, the tail regions ($n < 10$ th percentile and $n > 90$ th percentile) contain sparse data with substantial

gaps between consecutive observed sentence lengths. This sparsity, combined with near-zero local variance at individual data points, limits the models' ability to accurately predict extreme values. Consequently, systematic deviations occur in both tails: slight underestimation for very short sentences ($n < 10$) and occasional overestimation for very long sentences ($n > 50$). These patterns are consistent across all 21 languages and visible in both linear and log-log representations.

3 Discussion

3.1 Summary of Results

Our analysis of the scaling relationship between sentence length (n) and the second moment of degree distribution ($\langle k^2 \rangle$) across 21 languages reveals consistent non-linear patterns that fundamentally contradict random tree baselines. Model selection via AIC and BIC converges on Model 5 ($f(n) = an^be^{cn}$) or its extended variant Model 5+ ($f(n) = an^be^{cn} + d$) as optimal across all languages, with astronomical ΔAIC and ΔBIC values decisively rejecting the null hypothesis.

The fitted parameters exhibit remarkable stability: power-law exponents b cluster around 0.40 (range: 0.30–0.55), exponential coefficients c are consistently small and negative (-0.016 to -0.005), and scaling factors a vary between 1.5 and 2.6. This parametric consistency across typologically diverse languages, spanning isolating (Chinese), agglutinative (Turkish, Finnish), and fusional (Romance, Slavic) morphologies, points to universal syntactic constraints independent of specific grammatical implementations.

Model performance metrics quantify this consistency: R^2 values for Model 5/5+ range from 0.32 to 0.55 with coefficient of variation 14%, while RSE values span 0.51 to 0.83 with coefficient of variation 12%. The tight clustering of performance metrics across languages reinforces the universality of the underlying scaling relationship.

3.2 Rejection of the Null Hypothesis

The null model, predicting $\langle k^2 \rangle$ based on uniformly random labeled trees, fails catastrophically across all languages. Negative R^2 values (ranging from -0.88 to -3.76) indicate performance worse than predicting the mean, while RSS values exceed those of best alternatives by factors of 4 or 9. Information criteria reflect this failure with extreme ΔAIC and ΔBIC values: English exhibits $\Delta\text{AIC} = 1,562,501,238$, while more typical values like Arabic ($\Delta\text{AIC} = 955$) and Czech ($\Delta\text{AIC} = 1,279$) still represent overwhelming evidence against the null.

This systematic rejection demonstrates that natural language syntax imposes substantially stronger structural constraints than random tree generation. Real dependency structures follow predictable sub-linear scaling patterns governed by power-law growth with exponential dampening, rather than the quadratic behavior predicted by random baselines. The universal deviation from randomness suggests fundamental cognitive or communicative pressures that shape syntactic organization across all human languages.

3.3 Heteroskedasticity and Model Validity

White's test revealed pervasive heteroskedasticity across virtually all model-language combinations (Table 2) with p-values typically below 10^{-10} , systematically violating the constant-variance assumption underlying standard likelihood calculations. This necessitated a heteroskedasticity-robust framework combining LOWESS variance estimation (bandwidth = 0.3), modified log-likelihood incorporating observation-specific variances, and corrected AIC/BIC accounting for effective degrees of freedom ($k_{\text{LOWESS}} \approx 0.3N$).

This approach ensures that the squared residuals are appropriately weighted by their estimated variances, giving less influence to high-variance observations while penalizing models that produce highly variable predictions. The corrected information criteria maintain valid relative model rankings despite the heteroskedastic error structure.

3.4 Model Selection: AIC versus BIC

Comparison of AIC and BIC model selection reveals a systematic divergence reflecting the fundamental trade-off between predictive accuracy and parsimony. AIC consistently favors Model 5 or Model 5+ for most languages (e.g., English, French, German selecting Model 5; Arabic, Czech, Italian, Polish selecting Model 5+). BIC, with its stronger complexity penalty proportional to $\log(N) \approx 6.91$ rather than AIC's fixed penalty of 2, often shifts preference toward simpler alternatives.

For several languages, BIC selects the same model family but favors the version without the offset parameter (e.g., Model 5 over Model 5+). In some cases, BIC supports substantially simpler models: Chinese shows near-equivalence between Model 5 and Model 4+ (logarithmic with offset) under BIC ($\Delta\text{BIC} = 0.00$ vs. 0.05), while Korean selects Model 4+ under both criteria.

Despite these differences, both criteria decisively reject the null model and converge on a narrow set of best-performing alternatives. This consistency suggests that Model 5/5+ captures genuine structural properties of syntactic scaling, while the AIC-BIC divergence primarily reflects different philosophies regarding the parsimony-complexity trade-off rather than fundamental disagreement about model adequacy.

3.5 Expected Model Behaviour

Visual inspection of the fitted curves (Appendix B) reveals that model performance varies systematically as a function of data density across sentence length ranges. This variation reflects the underlying distribution of observations rather than inherent model inadequacy.

3.5.1 Impact of Data Distribution on Model Fit

The distribution of the data directly impacts model fitting. The central 80% of the data (from the 10th to 90th percentiles of sentence length) is dense and relatively uniform, providing robust constraints for parameter estimation. In this region, the models achieve excellent fit quality, accurately capturing both mean trends and variance.

In contrast, the tail regions (below the 10th and above the 90th percentiles) are sparse, which presents challenges. The wide dispersion of data points offers weak constraints, and the near zero local variance makes it difficult to distinguish between functional forms. Consequently, systematic deviations in the tails are an expected artifact of fitting continuous models to non-uniformly distributed data.

- For very short sentences ($n < 10\text{th percentile}$), models tend to slightly underestimate $\langle k^2 \rangle$, possibly due to discrete combinatorial effects at small sentence sizes.
- For very long sentences ($n > 90\text{th percentile}$), models occasionally overestimate $\langle k^2 \rangle$, which may reflect sampling variability or increased structural regularity.

3.5.2 Overall Model Assessment

The models are optimized via heteroskedasticity-corrected maximum likelihood to minimize weighted residuals across the entire dataset, naturally prioritizing the information-rich central region where predictive accuracy is most reliable and statistically justified. The excellent fit of Model 5/5+ in the central 80% range, combined with its superior handling of the tails and consistent parameter estimates across 21 typologically diverse languages (Table 8), demonstrates that it successfully captures the fundamental scaling relationship between sentence length and network complexity in natural language syntax.

It is important to recognize that "best model" refers only to optimal performance within our specific ensemble of candidate functions. The true data-generating process may involve additional complexity, such as regime-switching behavior at extreme sentence lengths or interactions with unmeasured variables, not captured by any single continuous parametric function. Nevertheless, the consistent superior performance of Model 5/5+ across all languages provides strong evidence for its validity as a description of typical syntactic scaling patterns.

3.6 Model Comparison: 5/5+ vs. 4+

While several models demonstrate strong performance, a detailed comparison between Model 4+ and Model 5/5+ offers the clearest insight into the underlying data structure. Model 4+, defined as $f(n) = a \cdot \log(n) + d$, assumes a simple logarithmic growth. This functional form is highly effective when the relationship between sentence length and complexity follows a saturating trend, making it the best-performing alternative to Model 5/5+ in many cases. It excels when the data in the right tail (long sentences) continues the trend established by the central data cloud.

However, in most languages, the extreme right tail exhibits more complex behavior. For very long sentences, the average squared degree, $\langle k^2 \rangle$, often plateaus or even decreases, deviating from the simple logarithmic curve. This is where Model 4+ fails and the superior flexibility of Model 5/5+ becomes apparent.

Model 5/5+, defined as $f(n) = a \cdot n^b \cdot e^{c \cdot n} (+d)$, combines polynomial and exponential components. This

allows it to capture a much wider range of behaviors. The exponential term, $e^{c \cdot n}$, is particularly crucial, as a negative value for the parameter c allows the model to capture the downturn in complexity observed at extreme sentence lengths. This ability to accurately model the non-monotonic trend in the heavy tails, while maintaining an excellent fit in the central region, establishes Model 5/5+ as the most robust and accurate choice across the languages.

A notable exception is observed for the Korean language (Figure ??), where Model 4+ outperforms Model 5/5+. This is likely because the tails of the Korean sentence length distribution do not exhibit the same complex downturn, allowing the simpler logarithmic model to provide a more parsimonious and accurate fit.

3.7 Cross-Linguistic Universality and Variation

The most striking finding is extensive cross-linguistic similarity despite testing 21 typologically diverse languages. Consistent model rankings, similar parameter values (power-law exponents 0.30–0.55, exponential coefficients -0.016 to -0.005), comparable R^2 values (0.32–0.55), and universal strong departures from random baselines suggest that $\langle k^2 \rangle$ scaling reflects deep cognitive or communicative constraints transcending specific grammatical systems.

The quantitative similarity in model performance reinforces this universality. Not only do the same models dominate across languages, but they achieve remarkably similar R^2 values and RSE values. This consistency indicates that sentence length and network complexity relationships are governed by similar principles across all languages tested, with cross-linguistic differences representing minor perturbations around a universal baseline rather than fundamentally different scaling regimes.

Nevertheless, subtle language-specific variations emerge. Korean and Turkish exhibit consistently lower mean $\langle k^2 \rangle$ values (4.73, 4.97) compared to other languages (typically 5.2–6.0), potentially reflecting agglutinative morphology and head-final word order. Japanese shows the highest mean sentence length ($\mu_n = 28.79$) and mean $\langle k^2 \rangle = 6.00$, possibly due to particle tokenization in dependency annotation. Chinese displays unique BIC preference for logarithmic Model 4+, potentially relating to isolating morphology and relatively rigid word order.

While statistically significant, these variations are quantitatively small compared to shared cross-linguistic patterns. Possible explanations for universality include processing efficiency constraints imposed by working memory limitations, balance between information density and parseability, or evolutionary convergence toward similar structural optima. The universal sub-linear scaling with exponential dampening suggests that syntactic complexity increases with sentence length but at a decreasing rate, possibly reflecting cognitive bounds on sustainable complexity growth.

3.8 Conclusions

This study demonstrates that the second moment of degree distribution in syntactic dependency trees follows consistent, predictable patterns across 21 typologically diverse languages. Complex functional forms combining power-law and exponential features (Model 5/5+) provide optimal descriptions, vastly outperforming random baseline models. The null hypothesis of random tree scaling is decisively rejected with overwhelming statistical evidence across all languages.

Pervasive heteroskedasticity necessitated methodological adaptations through LOWESS variance estimation and corrected information criteria, ensuring valid model comparison despite non-constant error variance. Both AIC and BIC converge on the same model family despite differing parsimony-complexity trade-offs, with BIC systematically favoring simpler variants.

The best models achieve excellent fit in the central 80% of data where observations are dense, though systematic deviations occur in sparse tail regions. Moderate R^2 values (0.32–0.55) indicate substantial unexplained variance, suggesting that factors beyond sentence length, like genre, discourse structure or annotation conventions, influence network complexity.

Despite minor cross-linguistic variations, the remarkable similarity in scaling behavior, parameter values, and model preferences points to universal cognitive or communicative constraints shaping human syntactic structure. The consistent selection of Model 5/5+ across diverse language families, combined with stable parameter estimates, provides strong evidence that this functional form captures fundamental aspects of syntactic scaling. However, the moderate fit quality and tail-region deviations suggest that additional factors beyond simple sentence length influence syntactic network complexity, warranting further investigation into genre effects, discourse constraints, and higher-order structural features.

4 Methods

In our experiments, we used the following python packages:

- **Numpy** for fast operations over arrays and for a polynomial regression handling.
- **Pandas** for the management of the data-frames.
- **Os** for the functioning of the `.py` files
- **NetworkX** for the graphs implementation.
- **Scipy** for the optimization of the proposed models.
- **Statsmodels** for the implementation of complex statistical elements.

4.1 Data Preparation

To generate the dependency tree metrics we proceeded as follows:

1. Retrieved the CoNLL-U files available from the past laboratory experiments (Laboratory 2).
2. Built the syntactic dependency trees for each sentence in the PUD tree-bank thanks to the CoNLL-U files in each folder, which permit to extract the syntactic dependencies of each sentence for each language (each file is contained in the folder `data \PUD`). To do this we created a `parse_ud_conllu` function that, given the structure of the CoNLL-U file, permits to extract all the useful information about the dependencies such as the parent-child relations and all sentence tokens.
3. Thanks to the python function previously mentioned, we extracted each token as a node and each parent-child relation as a directed edge and then constructed, thanks to the python library **NetworkX**, the dependency tree structure for each sentence in each language.
4. For each sentence in each language, we computed k_i , d_i and q .
5. For each sentence, we computed $\langle k^2 \rangle$, $\langle d \rangle$, q .
6. We stored these metrics in CSV files with the naming convention `lang_dependency_tree_metrics.csv`.
7. We validated the computed metrics against theoretical bounds:

$$\langle k^2 \rangle : 4 - \frac{6}{n} \leq \langle k^2 \rangle \leq n - 1 \quad (1)$$

$$\langle d \rangle : \frac{1}{4(n-1)}(n\langle k^2 \rangle + q) \leq \langle d \rangle \leq \frac{1}{4(n-1)}(3(n-1)^2 + 1 - n \bmod 2) \quad (2)$$

With these sentence-level metrics, we extracted a summary table containing, for each language, the number of sentences N , the μ and σ of n and $\langle k^2 \rangle$ (see Table 1).

4.2 The Ensemble of Models

We fit an ensemble of 11 models to describe the relationship between $\langle k^2 \rangle$ and n across all languages.

4.2.1 Model Definitions

The complete ensemble consists of :

$$\text{Model 0: } f(n) = \left(1 - \frac{1}{n}\right) \left(5 - \frac{6}{n}\right) \quad (3)$$

$$\text{Model 1: } f(n) = \left(\frac{n}{2}\right)^b \quad (4)$$

$$\text{Model 2: } f(n) = a \cdot n^b \quad (5)$$

$$\text{Model 3: } f(n) = a \cdot e^{c \cdot n} \quad (6)$$

$$\text{Model 4: } f(n) = a \cdot \log(n) \quad (7)$$

$$\text{Model 5: } f(n) = a \cdot n^b \cdot e^{c \cdot n} \quad (8)$$

The Extended Models (with offset parameter d) are defined as:

$$\text{Model 1+}: f(n) = \left(\frac{n}{2}\right)^b + d \quad (9)$$

$$\text{Model 2+}: f(n) = a \cdot n^b + d \quad (10)$$

$$\text{Model 3+}: f(n) = a \cdot e^{c \cdot n} + d \quad (11)$$

$$\text{Model 4+}: f(n) = a \cdot \log(n) + d \quad (12)$$

$$\text{Model 5+}: f(n) = a \cdot n^b \cdot e^{c \cdot n} + d \quad (13)$$

4.2.2 Initial Parameter Estimation

To ensure robust convergence of the non-linear optimization, we derive analytical initial guesses for each model's parameters by transforming the problem into linear regression in appropriate spaces.

Model 1

Taking logarithms on both sides:

$$\log(y) = b \cdot \log\left(\frac{n}{2}\right) = b \cdot \log(n/2) \quad (14)$$

We perform linear regression of $\log(y)$ against $\log(n/2)$ to obtain b_{init} .

Model 2

Similarly:

$$\log(y) = \log(a) + b \cdot \log(n) \quad (15)$$

Linear regression yields: b_{init} from the slope and $a_{\text{init}} = \exp(\text{intercept})$.

Model 3

Taking logarithms:

$$\log(y) = \log(a) + c \cdot n \quad (16)$$

We fit $\log(y)$ versus n to obtain c_{init} and $a_{\text{init}} = \exp(\text{intercept})$.

Model 4

This is already linear in the transformed space:

$$y = a \cdot \log(n) \quad (17)$$

Direct linear regression of y against $\log(n)$ gives a_{init} .

Model 5

Taking logarithms:

$$\log(y) = \log(a) + b \cdot \log(n) + c \cdot n \quad (18)$$

We perform multiple linear regression with predictors $\log(n)$ and n to estimate all three parameters:

$$a_{\text{init}} = \exp(\beta_0), \quad b_{\text{init}} = \beta_1, \quad c_{\text{init}} = \beta_2$$

Extended Models

For models with the d parameter, we initialize the base parameters using the estimates from their corresponding base models and set $d_{\text{init}} = 0$.

4.2.3 Non-linear Least Squares Fitting

All models (except the null) are fitted using the Levenberg-Marquardt [1] algorithm via `curve_fit` function. This function minimizes the residual sum of squares:

$$\text{RSS} = \sum_{i=1}^N (y_i - f(n_i; \boldsymbol{\theta}))^2 \quad (19)$$

where $\boldsymbol{\theta}$ represents the parameter vector for each model.

The standard `curve_fit` implementation assumes homoskedastic errors (constant σ^2 across all observations):

$$\epsilon_i \sim \mathcal{N}(0, \sigma^2), \quad i = 1, \dots, N \quad (20)$$

4.3 The Heteroskedastic Problem

After fitting all models using ordinary least squares (OLS), we must validate a critical assumption underlying standard information criteria: homoskedasticity (constant error variance across all observations).

The standard likelihood function used in AIC and BIC computation assumes:

$$\epsilon_i \sim \mathcal{N}(0, \sigma^2), \quad \text{for all } i = 1, \dots, N \quad (21)$$

where σ^2 is constant. Under this assumption, the log-likelihood is:

$$\log \mathcal{L}(\boldsymbol{\theta}) = -\frac{N}{2} \log(2\pi) - \frac{N}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^N (y_i - \hat{y}_i)^2 = \quad (22)$$

However, empirical analysis of our fitted models reveals systematic patterns in the residuals that violate this assumption. To formally test for heteroskedasticity, we employ White's test for heteroskedasticity.

4.3.1 White's Test for Heteroskedasticity

White's test is a general specification test that detects heteroskedasticity of unknown form without requiring specific assumptions about the variance structure. The test procedure is:

1. Fit the model and obtain residuals: $\hat{\epsilon}_i = y_i - \hat{y}_i$
2. Regress the squared residuals $\hat{\epsilon}_i^2$ on the fitted values $\hat{y}_i$ and their squares:

$$\hat{\epsilon}_i^2 = \alpha_0 + \alpha_1 \hat{y}_i + \alpha_2 \hat{y}_i^2 + \nu_i \quad (23)$$

3. Compute the test statistic:

$$\text{LM} = N \cdot R^2 \quad (24)$$

where R^2 is the coefficient of determination from the auxiliary regression, and N is the number of observations.

4. Under the null hypothesis of homoskedasticity ($H_0 : \alpha_1 = \alpha_2 = 0$), the test statistic follows asymptotically:

$$\text{LM} \sim \chi^2(2) \quad (25)$$

If the p -value is below a significance level (typically $\alpha = 0.05$), we reject the null hypothesis and conclude that heteroskedasticity is present.

We performed White's test on the 11 models for each of the 21 languages. For all languages, the test yields extremely small p -values (typically $p < 10^{-10}$), providing overwhelming evidence against the homoskedasticity assumption. This systematic heteroskedasticity manifests as increasing variance with larger fitted values.

The presence of heteroskedasticity has several consequences:

- OLS parameter estimates $\hat{\boldsymbol{\theta}}$ remain unbiased and consistent, but are no longer fully efficient (they do not achieve minimum variance).
- Standard errors computed under homoskedasticity are incorrect, leading to invalid confidence intervals and hypothesis tests.
- Information criteria (AIC, BIC) computed with the standard homoskedastic likelihood are biased, potentially leading to incorrect model rankings.

Given this pervasive heteroskedasticity, we must derive heteroskedasticity-robust versions of the log-likelihood and information criteria to ensure valid model comparison. This requires estimating the variance function $\sigma_i^2 = \text{Var}(\epsilon_i | \hat{y}_i)$ and incorporating it into our statistical framework.

4.3.2 LOWESS Smoothing for Variance Estimation

To correct for heteroskedasticity, we must estimate observation-specific variances σ_i^2 as a function of fitted values. We employ LOWESS (Locally Weighted Scatterplot Smoothing) [2], a non-parametric regression technique that estimates the variance function without imposing a parametric form.

The LOWESS procedure estimates $\hat{\sigma}_i^2$ through the following steps:

1. Compute residuals from the fitted model:

$$\hat{\epsilon}_i = y_i - \hat{y}_i = y_i - f(n_i; \hat{\theta}) \quad (26)$$

2. Calculate squared residuals as raw variance proxies:

$$v_i = \hat{\epsilon}_i^2 \quad (27)$$

3. For each observation i , identify a neighborhood of points based on proximity in $\hat{y}_i$ space. With bandwidth parameter $\text{frac} = 0.3$, we use the $k = \lfloor 0.3 \times N \rfloor$ nearest neighbors.

4. Assign weights to neighbors using the tricube weight function:

$$w_{ij} = \begin{cases} \left(1 - \left|\frac{\hat{y}_i - \hat{y}_j}{d_i}\right|^3\right)^3 & \text{if } |\hat{y}_i - \hat{y}_j| \leq d_i \\ 0 & \text{otherwise} \end{cases} \quad (28)$$

where d_i is the distance to the k -th nearest neighbor. This weight function gives maximum weight to the target point and smoothly decreases to zero at the neighborhood boundary.

5. Fit a weighted linear regression at each point to predict v_j from $\hat{y}_j$:

$$\min_{\alpha_i, \beta_i} \sum_{j=1}^N w_{ij} (v_j - (\alpha_i + \beta_i \hat{y}_j))^2 \quad (29)$$

6. The estimated variance at point i is:

$$\hat{\sigma}_i^2 = \alpha_i + \beta_i \hat{y}_i \quad (30)$$

7. Enforce numerical stability by setting a minimum variance threshold:

$$\hat{\sigma}_i^2 = \max(\hat{\sigma}_i^2, 10^{-10}) \quad (31)$$

After LOWESS estimation our model can be summarized as:

$$y_i = f(n_i; \theta) + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \hat{\sigma}_i^2) \quad (32)$$

where $\hat{\sigma}_i^2$ are now known observation-specific variances estimated from the data.

This transformation is crucial: while the errors remain non-constant across observations, their variances are now known quantities rather than unknown parameters. Conditional on the fitted values $\hat{y}_i$, we can treat each observation as having a known, fixed variance, effectively converting the heteroskedastic problem into a weighted least squares framework for likelihood computation.

The choice of $\text{frac} = 0.3$ represents a trade-off between bias and variance:

- Smaller values yield more flexible variance estimates but risk overfitting to noise in squared residuals.
- Larger values produce smoother estimates but may fail to capture genuine variance patterns.

The LOWESS smoother introduces additional model complexity. The effective degrees of freedom is given by:

$$k_{\text{LOWESS}} = \text{tr}(\mathbf{S}) \quad (33)$$

where $\mathbf{S}$ is the smoother matrix. An upper bound for the effective degrees of freedom of LOWESS with bandwidth parameter frac is $k_{\text{LOWESS}} \leq N \cdot \text{frac}$. Using the chosen value of frac , we can approximate as follows:

$$k_{\text{LOWESS}} \approx N \cdot \text{frac} = N \cdot 0.3 = 0.3N \quad (34)$$

Thus, the total number of parameters for model comparison is:

$$k_{\text{total}} = k_{\text{model}} + k_{\text{LOWESS}} = k_{\text{model}} + 0.3N \quad (35)$$

where k_{model} represents the structural parameters of the regression function.

4.3.3 Heteroskedastic Log-Likelihood

Using the LOWESS-estimated variances $\hat{\sigma}_i^2$, we can now compute a heteroskedasticity-corrected log-likelihood that properly accounts for the non-constant error variance structure.

The likelihood function for a single observation under heteroskedastic normal errors is:

$$L_i(\boldsymbol{\theta}) = \frac{1}{\sqrt{2\pi\hat{\sigma}_i^2}} \exp\left(-\frac{(y_i - \hat{y}_i)^2}{2\hat{\sigma}_i^2}\right) \quad (36)$$

The total likelihood for all N observations is the product:

$$L(\boldsymbol{\theta}) = \prod_{i=1}^N L_i(\boldsymbol{\theta}) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi\hat{\sigma}_i^2}} \exp\left(-\frac{(y_i - \hat{y}_i)^2}{2\hat{\sigma}_i^2}\right) \quad (37)$$

Taking the natural logarithm, we obtain the heteroskedasticity-corrected log-likelihood:

$$\log \mathcal{L}_{\text{hetero}}(\boldsymbol{\theta}) = \sum_{i=1}^N \log L_i(\boldsymbol{\theta}) \quad (38)$$

$$= \sum_{i=1}^N \left[-\frac{1}{2} \log(2\pi) - \frac{1}{2} \log(\hat{\sigma}_i^2) - \frac{(y_i - \hat{y}_i)^2}{2\hat{\sigma}_i^2} \right] \quad (39)$$

$$= -\frac{N}{2} \log(2\pi) - \frac{1}{2} \sum_{i=1}^N \log(\hat{\sigma}_i^2) - \frac{1}{2} \sum_{i=1}^N \frac{(y_i - \hat{y}_i)^2}{\hat{\sigma}_i^2} \quad (40)$$

This formulation differs from the standard homoskedastic log-likelihood in two key ways:

1. The squared residuals $(y_i - \hat{y}_i)^2$ are weighted by their respective variances $\hat{\sigma}_i^2$, giving less influence to observations with high estimated variance (high uncertainty).
2. The term $\sum_{i=1}^N \log(\hat{\sigma}_i^2)$ penalizes models that produce highly variable predictions, even if they achieve small residuals in some regions.

This corrected log-likelihood provides the foundation for computing heteroskedasticity information criteria.

4.3.4 Model Selection with Heteroskedasticity-Corrected Information Criteria

With the heteroskedasticity-corrected log-likelihood $\log \mathcal{L}_{\text{hetero}}(\boldsymbol{\theta})$, we can now compute modified AIC and BIC that properly account for both the model structural parameters and the complexity introduced by variance estimation.

The total effective number of parameters now includes:

$$k_{\text{total}} = k_{\text{model}} + k_{\text{LOWESS}} \quad (41)$$

where:

- k_{model} = number of parameters in $f(n; \boldsymbol{\theta})$
- $k_{\text{LOWESS}} \approx 0.3N$ = effective degrees of freedom for LOWESS variance estimation

The Akaike Information Criterion penalizes model complexity linearly:

$$\text{AIC}_{\text{hetero}} = 2k_{\text{total}} - 2 \log \mathcal{L}_{\text{hetero}}(\hat{\boldsymbol{\theta}}) \quad (42)$$

Expanding:

$$\text{AIC}_{\text{hetero}} = 2(k_{\text{model}} + 0.3N) - 2 \log \mathcal{L}_{\text{hetero}}(\hat{\boldsymbol{\theta}}) \quad (43)$$

The Bayesian Information Criterion applies a stronger penalty that depends on sample size:

$$\text{BIC}_{\text{hetero}} = k_{\text{total}} \cdot \log(N) - 2 \log \mathcal{L}_{\text{hetero}}(\hat{\theta}) \quad (44)$$

Expanding:

$$\text{BIC}_{\text{hetero}} = (k_{\text{model}} + 0.3N) \cdot \log(N) - 2 \log \mathcal{L}_{\text{hetero}}(\hat{\theta}) \quad (45)$$

For both criteria, lower values indicate better models.

This adjustment ensures that the flexibility gained from modeling observation-specific variances is appropriately penalized in model comparison, preventing overfitting and providing valid relative model rankings despite the violation of homoskedasticity.

4.3.5 Model Performance Metrics: R^2 and RSE

To evaluate and compare model performance beyond information criteria, we computed two standard regression diagnostics: the coefficient of determination (R^2) and the residual standard error (RSE).

Coefficient of Determination (R^2) The coefficient of determination quantifies the proportion of variance in the observed data explained by the model. It is defined as:

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}} \quad (46)$$

where RSS is the residual sum of squares:

$$\text{RSS} = \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (47)$$

and TSS is the total sum of squares:

$$\text{TSS} = \sum_{i=1}^N (y_i - \bar{y})^2 \quad (48)$$

where $\bar{y} = \frac{1}{N} \sum_{i=1}^N y_i$ is the sample mean of the observed values.

R^2 ranges from $-\infty$ to 1, where:

- $R^2 = 1$ indicates perfect fit (all variance explained)
- $R^2 = 0$ indicates the model performs no better than predicting the mean
- $R^2 < 0$ indicates the model performs worse than a horizontal line at the mean

Negative R^2 values can occur for non-linear models when the model's predictions are systematically poor, as RSS exceeds TSS. This is commonly observed for the null model, which imposes a specific functional form derived from random tree theory rather than fitting to the data.

Residual Standard Error (RSE) The residual standard error provides an estimate of the standard deviation of the prediction errors, expressed in the original units of the response variable $\langle k^2 \rangle$.

For models fitted via ordinary least squares under the homoskedasticity assumption, RSE is computed as:

$$\text{RSE}_{\text{homo}} = \sqrt{\frac{\text{RSS}}{N - k_{\text{model}}}} \quad (49)$$

where k_{model} is the number of parameters in the regression function (excluding the variance parameter σ^2).

However, as established in Section 3.4, our data exhibit pervasive heteroskedasticity, necessitating a corrected formulation. Under the heteroskedastic framework with LOWESS-estimated variances $\hat{\sigma}_i^2$, the effective degrees of freedom must account for both the model parameters and the smoothing procedure:

$$\text{RSE}_{\text{hetero}} = \sqrt{\frac{\text{RSS}}{N - k_{\text{total}}}} \quad (50)$$

where the total effective degrees of freedom is:

$$k_{\text{total}} = k_{\text{model}} + k_{\text{LOWESS}} \quad (51)$$

Both R^2 and RSE provide complementary perspectives on model performance: R^2 offers a dimensionless, relative measure of explanatory power, while RSE provides an absolute measure of prediction accuracy in the original measurement scale. Together with the information criteria (AIC, BIC), these metrics enable comprehensive model evaluation and selection.

References

- [1] D. W. Marquardt, *An Algorithm for Least-Squares Estimation of Nonlinear Parameters*, Journal of the Society for Industrial and Applied Mathematics, 11(2):431–441, 1963.
- [2] W. S. Cleveland, *Robust Locally Weighted Regression and Smoothing Scatterplots*, Journal of the American Statistical Association, 74(368):829–836, 1979.

A Results tables

Table 2: White's Test for Heteroskedasticity: p-values

	Null	Model 1	Model 2	Model 3	Model 4	Model 5	Model 1+	Model 2+	Model 3+	Model 4+	Model 5+
arabic	2.87×10^{-15}	2.65×10^{-105}	2.53×10^{-4}	5.93×10^{-36}	1.48×10^{-6}	5.65×10^{-1}	4.80×10^{-9}	1.27×10^{-1}	1.74×10^{-27}	1.27×10^{-1}	5.13×10^{-1}
chinese	1.40×10^{-17}	2.68×10^{-90}	5.57×10^{-1}	1.06×10^{-11}	4.09×10^{-6}	8.91×10^{-1}	6.11×10^{-2}	8.85×10^{-1}	4.76×10^{-8}	8.85×10^{-1}	8.44×10^{-1}
czech	1.86×10^{-19}	5.17×10^{-100}	2.75×10^{-2}	1.12×10^{-27}	6.83×10^{-5}	7.17×10^{-1}	1.51×10^{-5}	6.21×10^{-1}	3.97×10^{-19}	6.21×10^{-1}	5.86×10^{-1}
english	2.77×10^{-5}	1.80×10^{-80}	6.98×10^{-2}	1.16×10^{-9}	3.57×10^{-3}	5.49×10^{-1}	7.10×10^{-3}	2.33×10^{-1}	2.52×10^{-7}	2.33×10^{-1}	5.53×10^{-1}
finnish	1.31×10^{-16}	7.25×10^{-74}	6.51×10^{-1}	2.26×10^{-8}	2.58×10^{-1}	8.32×10^{-2}	5.36×10^{-1}	2.40×10^{-1}	9.72×10^{-4}	2.40×10^{-1}	3.83×10^{-2}
french	5.68×10^{-8}	9.14×10^{-98}	1.65×10^{-2}	1.11×10^{-14}	2.01×10^{-9}	2.56×10^{-1}	3.42×10^{-4}	1.24×10^{-1}	8.12×10^{-12}	1.24×10^{-1}	2.84×10^{-1}
galician	2.80×10^{-15}	2.50×10^{-128}	2.85×10^{-9}	3.29×10^{-50}	2.40×10^{-13}	1.61×10^{-1}	3.86×10^{-17}	9.65×10^{-4}	2.77×10^{-41}	9.69×10^{-4}	3.87×10^{-1}
german	9.41×10^{-17}	7.18×10^{-106}	3.09×10^{-2}	9.06×10^{-19}	5.49×10^{-7}	1.92×10^{-1}	3.95×10^{-4}	2.23×10^{-1}	3.67×10^{-13}	2.23×10^{-1}	2.67×10^{-1}
hindi	1.40×10^{-12}	4.54×10^{-93}	4.13×10^{-2}	2.79×10^{-14}	2.56×10^{-7}	4.35×10^{-1}	1.40×10^{-3}	2.26×10^{-1}	8.05×10^{-11}	2.26×10^{-1}	5.53×10^{-1}
icelandic	1.86×10^{-9}	1.24×10^{-110}	6.98×10^{-2}	5.85×10^{-15}	3.10×10^{-9}	3.80×10^{-1}	2.01×10^{-3}	3.85×10^{-1}	1.22×10^{-10}	3.85×10^{-1}	5.51×10^{-1}
indonesian	7.44×10^{-8}	3.44×10^{-82}	1.29×10^{-1}	7.87×10^{-16}	6.66×10^{-3}	7.06×10^{-1}	3.15×10^{-3}	6.11×10^{-1}	4.08×10^{-11}	6.11×10^{-1}	5.96×10^{-1}
italian	4.23×10^{-14}	7.26×10^{-121}	6.55×10^{-13}	2.23×10^{-49}	5.22×10^{-16}	4.54×10^{-3}	7.94×10^{-21}	9.42×10^{-7}	4.44×10^{-42}	9.47×10^{-7}	1.44×10^{-1}
japanese	6.25×10^{-14}	5.05×10^{-116}	1.19×10^{-1}	1.31×10^{-13}	5.60×10^{-10}	7.42×10^{-1}	5.43×10^{-3}	4.45×10^{-1}	8.53×10^{-11}	4.45×10^{-1}	6.69×10^{-1}
korean	2.10×10^{-5}	6.17×10^{-104}	4.38×10^{-1}	1.97×10^{-8}	5.77×10^{-6}	5.05×10^{-1}	2.02×10^{-1}	4.91×10^{-1}	2.60×10^{-5}	4.91×10^{-1}	4.18×10^{-1}
polish	6.87×10^{-11}	1.17×10^{-107}	2.47×10^{-6}	8.53×10^{-36}	4.18×10^{-11}	3.59×10^{-2}	8.65×10^{-11}	1.38×10^{-3}	1.67×10^{-27}	1.38×10^{-3}	3.59×10^{-2}
portuguese	1.78×10^{-17}	3.65×10^{-121}	1.74×10^{-7}	1.17×10^{-47}	7.15×10^{-14}	2.18×10^{-1}	1.50×10^{-14}	4.89×10^{-3}	1.21×10^{-38}	4.90×10^{-3}	4.39×10^{-1}
russian	1.58×10^{-18}	3.69×10^{-113}	5.19×10^{-4}	1.99×10^{-37}	3.60×10^{-8}	7.36×10^{-1}	7.79×10^{-9}	1.83×10^{-1}	6.17×10^{-28}	1.83×10^{-1}	6.15×10^{-1}
spanish	2.60×10^{-16}	1.32×10^{-134}	1.08×10^{-19}	5.55×10^{-68}	1.70×10^{-18}	1.22×10^{-4}	7.36×10^{-31}	2.34×10^{-10}	3.05×10^{-59}	2.36×10^{-10}	2.31×10^{-2}
swedish	1.49×10^{-13}	1.27×10^{-107}	2.85×10^{-2}	1.54×10^{-16}	1.00×10^{-6}	3.46×10^{-1}	3.86×10^{-4}	2.72×10^{-1}	7.13×10^{-12}	2.72×10^{-1}	5.32×10^{-1}
thai	2.75×10^{-3}	2.51×10^{-76}	2.67×10^{-4}	1.84×10^{-22}	2.13×10^{-6}	3.10×10^{-2}	5.97×10^{-7}	1.16×10^{-2}	8.58×10^{-18}	1.16×10^{-2}	2.60×10^{-2}
turkish	3.68×10^{-12}	1.87×10^{-110}	3.06×10^{-5}	2.03×10^{-38}	2.26×10^{-7}	8.82×10^{-2}	3.49×10^{-10}	2.41×10^{-2}	2.11×10^{-28}	2.41×10^{-2}	6.02×10^{-2}

Table 3: Coefficient of Determination (R^2) for all models across languages. Values closer to 1 indicate better fit. Negative values indicate the model performs worse than a horizontal line at the mean.

Language	Null	Model 1	Model 2	Model 3	Model 4	Model 5	Model 1+	Model 2+	Model 3+	Model 4+	Model 5+
arabic	-0.939	-1.484	0.494	0.387	0.345	0.524	0.474	0.511	0.408	0.511	0.529
chinese	-2.108	-1.472	0.400	0.336	0.263	0.409	0.388	0.407	0.351	0.407	0.411
czech	-1.325	-1.269	0.480	0.379	0.388	0.512	0.460	0.498	0.404	0.498	0.516
english	-2.588	-1.717	0.296	0.229	0.127	0.319	0.282	0.306	0.242	0.306	0.323
finnish	-1.088	-0.844	0.449	0.363	0.408	0.475	0.432	0.464	0.390	0.464	0.477
french	-3.159	-2.584	0.362	0.282	0.059	0.390	0.346	0.373	0.295	0.373	0.394
galician	-2.253	-2.130	0.482	0.358	0.259	0.522	0.457	0.502	0.380	0.502	0.532
german	-2.818	-1.815	0.430	0.344	0.266	0.454	0.414	0.443	0.363	0.443	0.457
hindi	-2.680	-1.869	0.363	0.301	0.167	0.374	0.352	0.370	0.314	0.370	0.377
icelandic	-1.838	-1.929	0.395	0.321	0.187	0.414	0.381	0.405	0.337	0.405	0.416
indonesian	-1.342	-1.509	0.411	0.328	0.263	0.434	0.395	0.424	0.346	0.424	0.439
italian	-2.832	-2.384	0.439	0.328	0.181	0.479	0.416	0.456	0.347	0.456	0.488
japanese	-3.768	-2.127	0.301	0.239	0.075	0.318	0.290	0.309	0.250	0.309	0.322
korean	-0.883	-1.524	0.424	0.355	0.231	0.434	0.414	0.432	0.373	0.432	0.437
polish	-1.221	-1.652	0.464	0.368	0.302	0.495	0.445	0.479	0.390	0.479	0.498
portuguese	-2.330	-1.994	0.477	0.364	0.271	0.511	0.455	0.495	0.385	0.495	0.518
russian	-1.111	-1.397	0.504	0.397	0.382	0.536	0.484	0.522	0.422	0.522	0.540
spanish	-2.484	-2.460	0.500	0.380	0.230	0.539	0.477	0.520	0.401	0.520	0.548
swedish	-2.212	-1.730	0.426	0.347	0.270	0.445	0.411	0.437	0.366	0.437	0.447
thai	-0.892	-1.623	0.392	0.295	0.194	0.426	0.372	0.409	0.311	0.409	0.435
turkish	-0.525	-1.247	0.484	0.382	0.371	0.515	0.465	0.503	0.407	0.503	0.521

Table 4: Residual Standard Error (RSE) for all models across languages. Lower values indicate better predictive accuracy. RSE is measured in the same units as $\langle k^2 \rangle$.

Language	Null	Model 1	Model 2	Model 3	Model 4	Model 5	Model 1+	Model 2+	Model 3+	Model 4+	Model 5+
arabic	1.184	1.342	0.606	0.667	0.689	0.588	0.618	0.596	0.655	0.596	0.585
chinese	1.707	1.523	0.751	0.790	0.832	0.746	0.758	0.747	0.782	0.746	0.745
czech	1.491	1.474	0.706	0.772	0.766	0.685	0.720	0.694	0.757	0.694	0.682
english	1.887	1.643	0.837	0.876	0.931	0.824	0.845	0.832	0.869	0.831	0.822
finnish	1.652	1.554	0.850	0.914	0.880	0.831	0.863	0.838	0.895	0.838	0.829
french	1.665	1.547	0.653	0.693	0.793	0.639	0.661	0.648	0.687	0.648	0.637
galician	1.492	1.465	0.596	0.664	0.713	0.573	0.611	0.585	0.653	0.584	0.568
german	1.815	1.559	0.702	0.753	0.796	0.688	0.712	0.695	0.743	0.694	0.686
hindi	1.745	1.542	0.727	0.762	0.831	0.721	0.733	0.723	0.755	0.723	0.720
icelandic	1.498	1.523	0.693	0.734	0.803	0.683	0.700	0.687	0.726	0.687	0.681
indonesian	1.366	1.415	0.686	0.732	0.766	0.673	0.695	0.678	0.723	0.678	0.670
italian	1.635	1.537	0.627	0.685	0.756	0.604	0.639	0.617	0.676	0.617	0.599
japanese	1.872	1.517	0.718	0.749	0.825	0.709	0.724	0.714	0.744	0.714	0.708
korean	0.831	1.240	0.593	0.628	0.685	0.588	0.598	0.589	0.619	0.589	0.587
polish	1.276	1.396	0.628	0.682	0.716	0.610	0.639	0.619	0.671	0.619	0.608
portuguese	1.513	1.435	0.600	0.662	0.708	0.581	0.613	0.591	0.651	0.590	0.577
russian	1.303	1.389	0.632	0.697	0.705	0.612	0.645	0.621	0.683	0.621	0.610
spanish	1.413	1.409	0.536	0.597	0.665	0.515	0.548	0.526	0.587	0.526	0.510
swedish	1.662	1.533	0.703	0.750	0.793	0.692	0.712	0.697	0.740	0.697	0.691
thai	1.149	1.354	0.652	0.703	0.751	0.634	0.663	0.643	0.695	0.643	0.629
turkish	1.050	1.276	0.611	0.669	0.675	0.594	0.623	0.601	0.656	0.601	0.590

Table 5: Residual Sum of Squares (RSS) for all models across languages. Lower values indicate better fit.

Language	Null	Model 1	Model 2	Model 3	Model 4	Model 5	Model 1+	Model 2+	Model 3+	Model 4+	Model 5+
arabic	981.83	1258.35	256.48	310.75	331.74	241.33	266.64	247.65	298.98	247.65	238.40
chinese	2039.59	1622.06	394.10	435.89	483.95	387.76	401.55	388.93	426.02	388.93	386.25
czech	1556.41	1518.57	347.82	415.92	409.97	326.69	361.36	336.13	399.21	336.13	324.07
english	2492.03	1887.45	489.13	535.37	606.05	472.91	498.74	482.08	526.65	482.07	469.93
finnish	1910.85	1687.57	503.93	582.48	541.34	480.78	519.71	490.04	558.61	490.04	478.84
french	1940.08	1671.96	297.82	334.88	439.19	284.77	305.07	292.69	329.02	292.68	282.73
galician	1558.32	1499.54	248.18	307.75	355.25	228.86	260.21	238.45	297.06	238.45	224.25
german	2305.16	1699.50	343.87	395.97	442.96	329.63	354.01	336.39	384.48	336.38	327.57
hindi	2132.08	1662.15	369.01	405.01	482.84	362.44	375.40	364.69	397.67	364.68	361.04
icelandic	1571.38	1621.72	334.81	376.13	450.27	324.73	342.50	329.24	366.93	329.24	323.12
indonesian	1305.77	1398.67	328.38	374.39	410.63	315.67	337.46	320.81	364.53	320.81	312.47
italian	1870.19	1651.61	274.02	327.94	399.76	254.11	285.12	265.48	318.92	265.48	250.05
japanese	2453.09	1608.82	359.58	391.35	475.87	350.70	365.37	355.56	386.00	355.56	348.85
korean	483.85	1075.52	245.33	274.99	327.81	241.14	249.86	241.85	267.21	241.85	240.03
polish	1140.05	1361.39	275.18	324.46	358.20	259.23	284.71	267.30	313.36	267.29	257.54
portuguese	1602.15	1440.24	251.63	305.95	350.82	235.06	262.32	243.15	295.73	243.15	231.80
russian	1188.10	1349.03	279.06	339.42	347.66	261.41	290.56	269.14	325.12	269.13	258.97
spanish	1397.68	1388.08	200.40	248.69	308.77	185.13	210.00	192.76	240.16	192.76	181.16
swedish	1932.59	1642.29	345.25	392.65	439.02	333.97	354.22	338.67	381.22	338.67	332.47
thai	923.94	1280.79	296.92	344.50	393.75	280.13	306.49	288.58	336.48	288.58	275.68
turkish	772.00	1137.64	260.98	312.69	318.56	245.52	271.03	251.71	300.19	251.71	242.51

Table 6: Akaike Information Criterion (AIC) for all models across languages. Lower values indicate better models.

Language	Null	Model 1	Model 2	Model 3	Model 4	Model 5	Model 1+	Model 2+	Model 3+	Model 4+	Model 5+
arabic	3392.36	3400.91	2531.16	2702.41	2814.43	2449.77	2572.80	2495.08	2695.76	2493.00	2437.44
chinese	4142.16	3677.33	2683.69	2813.59	3002.72	2658.07	2720.50	2665.02	2795.92	2663.04	2658.16
czech	3822.78	3652.78	2598.88	2792.08	2888.72	2559.32	2635.69	2574.03	2741.92	2571.90	2544.23
english	1562504347.19	3920.17	3152.40	3243.18	3435.58	3109.01	3179.46	3151.80	3222.14	3149.74	3118.08
finnish	4077.51	3887.27	2974.10	3150.17	3171.87	2920.85	3012.92	2936.40	3093.25	2934.40	2911.94
french	4111.07	3571.94	2764.62	2810.55	3018.64	2632.92	2771.46	2769.45	2822.59	2767.43	2636.59
galician	3839.10	3481.25	2444.72	2638.76	2871.74	2322.76	2496.96	2399.28	2609.26	2397.25	2307.85
german	4263.31	3646.19	2630.29	2774.05	3025.35	2566.01	2669.53	2606.24	2768.86	2604.17	2558.92
hindi	4197.95	3682.92	2673.07	2791.38	3189.73	2640.01	2701.95	2652.66	2781.07	2650.43	2636.17
icelandic	3906.58	3604.80	2564.12	2704.07	2909.20	2523.10	2610.77	2538.50	2687.54	2536.49	2529.66
indonesian	3763.40	3527.86	2781.89	2976.13	3254.89	2677.91	2816.41	2744.64	2961.80	2742.59	2672.11
italian	156254003.52	3534.49	2321.05	2444.56	2768.81	2264.24	2357.99	2290.34	2439.98	2288.21	2257.76
japanese	156254303.61	3593.38	2676.27	2747.09	2936.00	2663.74	2690.32	2667.50	2740.90	2665.17	2651.65
korean	3118.56	3275.49	2359.27	2483.02	2708.05	2378.83	2373.81	2349.27	2454.37	2347.32	2373.83
polish	3583.19	3469.57	2436.37	2571.66	2857.24	2351.95	2460.86	2402.87	2565.25	2400.84	2335.80
portuguese	3870.07	3452.95	2338.94	2523.68	2747.06	2271.85	2399.84	2320.22	2496.54	2318.16	2269.03
russian	3555.63	3465.24	2410.45	2603.28	2669.57	2365.26	2449.92	2383.56	2563.83	2381.53	2371.00
spanish	1718753665.77	3300.50	1993.46	2144.19	2508.25	1954.25	2028.94	1979.96	2132.24	1977.94	1948.59
swedish	4096.01	3633.34	2650.70	2762.45	2977.12	2582.05	2678.72	2623.61	2732.67	2621.58	2584.80
thai	3481.40	3359.71	2693.72	2744.80	3121.45	2607.78	2704.60	2684.22	2760.46	2682.18	2580.10
turkish	3193.90	3293.44	2391.02	2565.98	2587.14	2334.20	2433.24	2358.74	2530.13	2356.71	2326.95

Table 7: ΔAIC values (difference from best model) for all models across languages. The best model for each language has $\Delta\text{AIC} = 0$. Values > 10 indicate essentially no support for the model.

Language	Null	Model 1	Model 2	Model 3	Model 4	Model 5	Model 1+	Model 2+	Model 3+	Model 4+	Model 5+
arabic	954.92	963.47	93.72	264.97	376.99	12.33	135.36	57.63	258.32	55.56	0.00
chinese	1484.09	1019.26	25.61	155.52	344.64	0.00	62.42	6.95	137.85	4.96	0.09
czech	1278.55	1108.55	54.65	247.86	344.50	15.09	91.46	29.81	197.69	27.67	0.00
english	1562501238.19	811.16	43.40	134.17	326.57	0.00	70.45	42.79	113.13	40.73	9.07
finnish	1165.56	975.33	62.16	238.23	259.92	8.90	100.98	24.46	181.31	22.46	0.00
french	1478.15	939.02	131.70	177.63	385.73	0.00	138.54	136.53	189.67	134.51	3.67
galician	1531.25	1173.40	136.87	330.91	563.89	14.92	189.11	91.43	301.42	89.40	0.00
german	1704.39	1087.26	71.36	215.13	466.43	7.08	110.60	47.31	209.94	45.25	0.00
hindi	1561.78	1046.75	36.90	155.21	553.56	3.84	65.78	16.48	144.90	14.26	0.00
icelandic	1383.48	1081.70	41.02	180.97	386.10	0.00	87.67	15.40	164.44	13.38	6.56
indonesian	1091.29	855.75	109.77	304.02	582.77	5.79	144.30	72.53	289.68	70.48	0.00
italian	156251745.76	1276.73	63.29	186.80	511.05	6.48	100.23	32.58	182.22	30.45	0.00
japanese	156251651.96	941.72	24.62	95.44	285.34	12.09	38.67	15.85	89.25	13.52	0.00
korean	771.24	928.18	11.96	135.70	360.73	31.51	26.49	1.96	107.05	0.00	26.51
polish	1247.39	1133.78	100.58	235.86	521.45	16.15	125.06	67.08	229.45	65.05	0.00
portuguese	1601.03	1183.92	69.91	254.65	478.02	2.81	130.81	51.19	227.51	49.13	0.00
russian	1190.37	1099.98	45.20	238.03	304.31	0.00	84.66	18.30	198.57	16.27	5.74
spanish	1718751717.17	1351.91	44.87	195.60	559.65	5.66	80.35	31.36	183.65	29.35	0.00
swedish	1513.95	1051.28	68.65	180.39	395.07	0.00	96.67	41.56	150.61	39.52	2.74
thai	901.30	779.61	113.62	164.70	541.35	27.68	124.50	104.12	180.36	102.08	0.00
turkish	866.95	966.49	64.07	239.04	260.20	7.26	106.30	31.80	203.19	29.76	0.00

Table 8: Fitted parameters for all models across languages.

Language	Model																							
	1		2		3		4	5			1+		2+			3+			4+		5+			
	<i>b</i>	<i>a</i>	<i>b</i>	<i>a</i>	<i>c</i>	<i>a</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>b</i>	<i>d</i>	<i>a</i>	<i>b</i>	<i>d</i>	<i>a</i>	<i>c</i>	<i>d</i>	<i>a</i>	<i>d</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	
Arabic	0.679	2.769	0.214	4.289	0.009	1.760	1.949	0.400	-0.009	0.423	2.584	9.20×10^3	1.22×10^{-4}	-9.20×10^3	9.02×10^3	5.87×10^{-6}	-9.01×10^3	1.126	1.908	4.06×10^3	4.41×10^{-4}	-9.05×10^{-6}	-4.06×10^3	
Chinese	0.704	3.018	0.210	4.645	0.009	1.889	2.398	0.329	-0.006	0.435	2.919	8.27×10^3	1.44×10^{-4}	-8.26×10^3	8.64×10^3	6.44×10^{-6}	-8.64×10^3	1.188	2.131	2.24×10^3	7.03×10^{-4}	-9.23×10^{-6}	-2.24×10^3	
Czech	0.722	2.749	0.236	4.356	0.011	1.890	1.840	0.459	-0.012	0.462	2.651	1.09×10^4	1.18×10^{-4}	-1.09×10^4	1.14×10^4	5.87×10^{-6}	-1.14×10^4	1.290	1.740	3.16×10^3	6.60×10^{-4}	-1.53×10^{-5}	-3.16×10^3	
English	0.716	3.321	0.187	4.905	0.008	1.939	2.304	0.378	-0.010	0.414	3.206	8.94×10^3	1.23×10^{-4}	-8.94×10^3	8.20×10^3	6.09×10^{-6}	-8.19×10^3	1.104	2.532	5.02×10^3	3.95×10^{-4}	-9.29×10^{-6}	-5.01×10^3	
Finnish	0.772	2.565	0.274	4.189	0.015	1.994	1.694	0.523	-0.016	0.514	2.526	8.82×10^3	1.68×10^{-4}	-8.82×10^3	1.26×10^4	7.24×10^{-6}	-1.26×10^4	1.485	1.396	1.21×10^3	1.91×10^{-3}	-4.83×10^{-5}	-1.21×10^3	
French	0.670	3.439	0.164	4.950	0.006	1.828	2.440	0.331	-0.007	0.374	3.247	6.34×10^3	1.51×10^{-4}	-6.34×10^3	8.70×10^3	4.26×10^{-6}	-8.69×10^3	0.957	2.769	3.68×10^3	4.68×10^{-4}	-9.43×10^{-6}	-3.68×10^3	
Galician	0.671	3.119	0.189	4.709	0.007	1.806	2.170	0.371	-0.008	0.400	2.960	1.29×10^4	8.30×10^{-5}	-1.29×10^4	1.13×10^4	3.84×10^{-6}	-1.13×10^4	1.069	2.307	5.91×10^3	3.11×10^{-4}	-6.42×10^{-6}	-5.91×10^3	
German	0.712	3.142	0.204	4.790	0.009	1.931	2.253	0.378	-0.009	0.433	3.060	9.28×10^3	1.28×10^{-4}	-9.28×10^3	1.20×10^4	4.57×10^{-6}	-1.20×10^4	1.188	2.255	2.95×10^3	6.45×10^{-4}	-1.29×10^{-5}	-2.95×10^3	
Hindi	0.682	3.251	0.185	4.851	0.007	1.857	2.559	0.303	-0.005	0.403	3.117	5.67×10^3	1.89×10^{-4}	-5.66×10^3	8.57×10^3	5.20×10^{-6}	-8.57×10^3	1.071	2.474	2.72×10^3	5.64×10^{-4}	-8.06×10^{-6}	-2.72×10^3	
Icelandic	0.720	3.062	0.200	4.523	0.010	1.889	2.308	0.356	-0.009	0.425	2.890	8.45×10^3	1.29×10^{-4}	-8.45×10^3	1.33×10^4	4.22×10^{-6}	-1.33×10^4	1.094	2.315	2.86×10^3	5.96×10^{-4}	-1.26×10^{-5}	-2.86×10^3	
Indonesian	0.706	2.883	0.212	4.385	0.010	1.833	2.064	0.394	-0.010	0.431	2.719	8.63×10^3	1.32×10^{-4}	-8.63×10^3	1.04×10^4	5.48×10^{-6}	-1.04×10^4	1.139	2.046	4.32×10^3	4.27×10^{-4}	-9.49×10^{-6}	-4.32×10^3	
Italian	0.677	3.279	0.179	4.853	0.007	1.840	2.245	0.368	-0.009	0.393	3.123	9.05×10^3	1.15×10^{-4}	-9.04×10^3	9.13×10^3	4.58×10^{-6}	-9.12×10^3	1.041	2.511	5.31×10^3	3.56×10^{-4}	-7.77×10^{-6}	-5.31×10^3	
Japanese	0.652	3.528	0.161	5.171	0.005	1.814	2.599	0.299	-0.005	0.367	3.380	7.74×10^3	1.26×10^{-4}	-7.74×10^3	8.04×10^3	4.05×10^{-6}	-8.04×10^3	0.974	2.802	3.38×10^3	4.85×10^{-4}	-7.68×10^{-6}	-3.38×10^3	
Korean	0.702	2.614	0.216	3.893	0.012	1.719	2.142	0.332	-0.007	0.423	2.328	6.20×10^3	1.66×10^{-4}	-6.19×10^3	8.33×10^3	7.17×10^{-6}	-8.33×10^3	1.026	1.933	2.17×10^3	6.29×10^{-4}	-1.02×10^{-5}	-2.16×10^3	
Polish	0.714	2.826	0.217	4.297	0.011	1.836	1.943	0.426	-0.012	0.437	2.653	6.88×10^3	1.67×10^{-4}	-6.87×10^3	8.05×10^3	7.48×10^{-6}	-8.05×10^3	1.147	1.992	2.25×10^3	8.51×10^{-4}	-2.07×10^{-5}	-2.24×10^3	
Portuguese	0.675	3.092	0.193	4.690	0.007	1.813	2.172	0.370	-0.008	0.406	2.943	8.82×10^3	1.23×10^{-4}	-8.82×10^3	9.99×10^3	4.54×10^{-6}	-9.99×10^3	1.088	2.266	4.92×10^3	3.69×10^{-4}	-7.35×10^{-6}	-4.92×10^3	
Russian	0.702	2.740	0.226	4.292	0.010	1.822	1.886	0.430	-0.011	0.444	2.594	8.89×10^3	1.36×10^{-4}	-8.88×10^3	9.70×10^3	6.24×10^{-6}	-9.70×10^3	1.206	1.812	3.67×10^3	5.25×10^{-4}	-1.15×10^{-5}	-3.67×10^3	
Spanish	0.671	3.143	0.184	4.683	0.007	1.794	2.225	0.357	-0.008	0.393	2.961	1.00×10^4	1.03×10^{-4}	-1.00×10^4	1.01×10^4	4.22×10^{-6}	-1.01×10^4	1.032	2.379	5.85×10^3	3.05×10^{-4}	-6.34×10^{-6}	-5.85×10^3	
Swedish	0.730	3.017	0.214	4.592	0.010	1.934	2.230	0.379	-0.009	0.445	2.913	8.44×10^3	1.42×10^{-4}	-8.43×10^3	1.03×10^4	5.98×10^{-6}	-1.03×10^4	1.200	2.147	2.22×10^3	8.21×10^{-4}	-1.64×10^{-5}	-2.22×10^3	
Thai	0.660	2.941	0.189	4.380	0.008	1.713	2.020	0.383	-0.009	0.390	2.701	9.38×10^3	1.06×10^{-4}	-9.38×10^3	9.13×10^3	4.75×10^{-6}	-9.13×10^3	0.997	2.206	5.46×10^3	3.25×10^{-4}	-7.46×10^{-6}	-5.46×10^3	
Turkish	0.720	2.579	0.237	4.007	0.013	1.796	1.776	0.455	-0.013	0.454	2.384	9.29×10^3	1.28×10^{-4}	-9.29×10^3	1.31×10^4	5.24×10^{-6}	-1.31×10^4	1.192	1.696	3.98×10^3	4.84×10^{-4}	-1.23×10^{-5}	-3.98×10^3	

Table 9: Absolute BIC values for all languages and models.

Language	Null	Model 1	Model 2	Model 3	Model 4	Model 5	Model 1+	Model 2+	Model 3+	Model 4+	Model 5+
english	1562505819.52	5397.40	4634.55	4725.32	4912.82	4596.06	4661.60	4638.85	4709.19	4631.88	4610.04
arabic	4864.69	4878.14	4013.30	4184.55	4291.67	3936.82	4054.95	3982.13	4182.81	3975.15	3929.40
czech	5295.11	5130.01	4081.02	4274.22	4365.96	4046.37	4117.83	4061.08	4228.97	4054.04	4036.19
german	5735.64	5123.42	4112.43	4256.19	4502.59	4053.06	4151.67	4093.29	4255.91	4086.31	4050.88
spanish	1718755138.09	4777.73	3475.61	3626.33	3985.48	3441.30	3511.08	3467.01	3619.29	3460.08	3440.55
finnish	5549.83	5364.51	4456.24	4632.31	4649.10	4407.90	4495.06	4423.45	4580.30	4416.55	4403.90
french	5583.39	5049.18	4246.76	4292.69	4495.88	4119.97	4253.60	4256.50	4309.64	4249.57	4128.54
galician	5311.42	4958.48	3926.86	4120.90	4348.97	3809.81	3979.10	3886.33	4096.31	3879.39	3799.80
hindi	5670.28	5160.16	4155.22	4273.52	4666.97	4127.06	4184.09	4139.71	4268.12	4132.57	4128.13
indonesian	5235.73	5005.09	4264.03	4458.28	4732.12	4164.95	4298.55	4231.69	4448.85	4224.73	4164.07
icelandic	5378.90	5082.04	4046.26	4186.21	4386.44	4010.15	4092.91	4025.55	4174.59	4018.63	4021.62
italian	156255475.85	5011.73	3803.19	3926.71	4246.05	3751.29	3840.13	3777.39	3927.03	3770.36	3749.72
japanese	156255775.94	5070.61	4158.42	4229.23	4414.23	4150.79	4172.47	4154.55	4227.95	4147.31	4143.61
korean	4590.89	4752.73	3841.42	3965.16	4185.29	3865.88	3855.95	3836.32	3941.42	3829.46	3865.79
polish	5055.51	4946.81	3918.52	4053.80	4334.48	3839.00	3943.00	3889.92	4052.30	3882.99	3827.75
portuguese	5342.39	4930.18	3821.08	4005.83	4224.29	3758.90	3881.98	3807.27	3983.59	3800.30	3760.99
russian	5027.96	4942.47	3892.60	4085.42	4146.80	3852.31	3932.06	3870.61	4050.88	3863.67	3862.95
swedish	5568.33	5110.57	4132.85	4244.59	4454.36	4069.10	4160.86	4110.66	4219.72	4103.72	4076.75
thai	4953.73	4836.95	4175.87	4226.94	4598.69	4094.83	4186.74	4171.27	4247.51	4164.33	4072.06
turkish	4666.22	4770.67	3873.16	4048.12	4064.38	3821.25	3915.38	3845.79	4017.18	3838.85	3818.90
chinese	5614.49	5154.56	4165.83	4295.73	4479.95	4145.12	4202.64	4152.07	4282.97	4145.18	4150.12

Table 10: ΔBIC values (difference from best model) for all models across languages. The best model for each language has $\Delta\text{BIC} = 0$. BIC penalizes complexity more strongly than AIC.

Language	Null	Model 1	Model 2	Model 3	Model 4	Model 5	Model 1+	Model 2+	Model 3+	Model 4+	Model 5+
arabic	935.29	948.74	83.90	255.15	362.27	7.42	125.54	52.73	253.41	45.74	0.00
chinese	1469.36	1009.44	20.70	150.61	334.83	0.00	57.51	6.95	137.85	0.05	5.00
czech	1258.92	1093.83	44.84	238.04	329.77	10.18	81.64	24.90	192.78	17.86	0.00
english	1562501223.46	801.34	38.49	129.26	316.76	0.00	65.54	42.79	113.13	35.82	13.98
finnish	1145.93	960.61	52.34	228.41	245.20	4.00	91.16	19.55	176.40	12.64	0.00
french	1463.43	929.21	126.79	172.72	375.91	0.00	133.63	136.53	189.67	129.60	8.58
galician	1511.62	1158.68	127.06	321.10	549.17	10.01	179.30	86.52	296.51	79.59	0.00
german	1684.76	1072.54	61.55	205.31	451.71	2.18	100.79	42.40	205.03	35.43	0.00
hindi	1543.22	1033.10	28.15	146.46	539.91	0.00	57.03	12.64	141.06	5.51	1.07
icelandic	1368.75	1071.89	36.11	176.06	376.29	0.00	82.76	15.40	164.44	8.48	11.47
indonesian	1071.66	841.02	99.96	294.21	568.05	0.89	134.48	67.62	284.78	60.66	0.00
italian	156251726.13	1262.01	53.48	176.99	496.33	1.57	90.42	27.67	177.31	20.64	0.00
japanese	156251632.33	927.00	14.81	85.62	270.62	7.18	28.86	10.94	84.34	3.70	0.00
korean	761.43	923.27	11.96	135.70	355.83	36.42	26.49	6.86	111.96	0.00	36.33
polish	1227.76	1119.05	90.76	226.05	506.72	11.24	115.24	62.17	224.55	55.23	0.00
portuguese	1583.50	1171.29	62.19	246.93	465.39	0.00	123.09	48.37	224.70	41.41	2.09
russian	1175.65	1090.17	40.29	233.12	294.50	0.00	79.75	18.30	198.57	11.37	10.65
spanish	1718751697.54	1337.19	35.06	185.78	544.93	0.75	70.53	26.46	178.74	19.53	0.00
swedish	1499.23	1041.47	63.74	175.49	385.25	0.00	91.76	41.56	150.61	34.61	7.65
thai	881.67	764.89	103.81	154.88	526.63	22.77	114.68	99.21	175.45	92.27	0.00
turkish	847.32	951.77	54.26	229.22	245.48	2.35	96.48	26.89	198.28	19.94	0.00

B Best Fit Models Plots

The best-fit plots for each of the 21 languages, illustrating the performance of the optimal model against the empirical data. The left panel displays the best model fit on a linear scale. It shows the raw scatter plot of the empirical data, the fitted curve and vertical dashed lines indicating the central 80% of the data based on sentence length. The right panel presents the same information on a log-log scale.

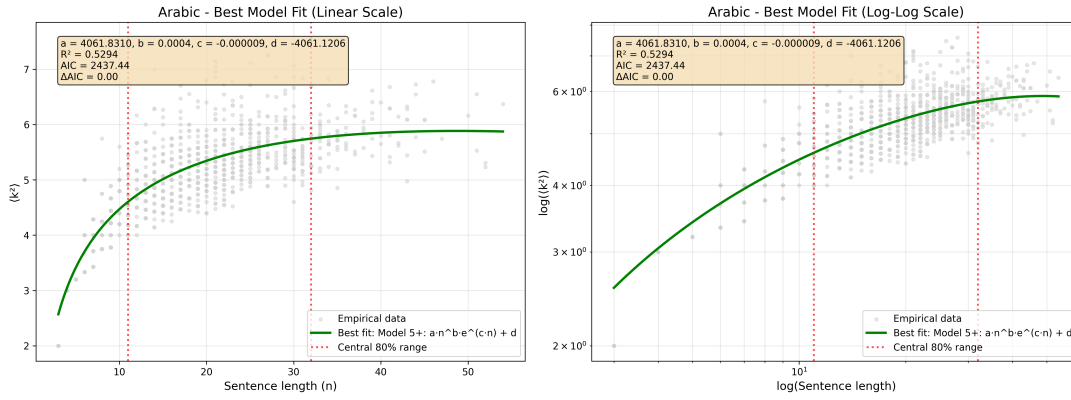


Figure 3: Arabic - Best fit model.

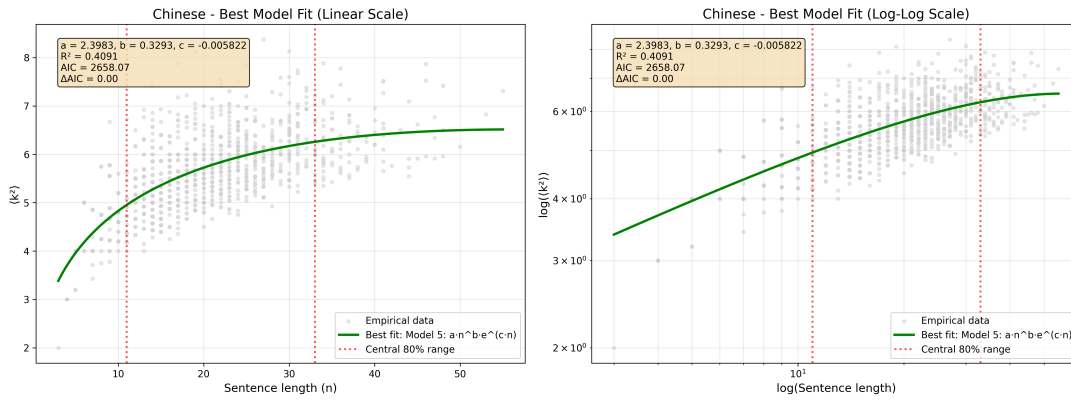


Figure 4: Chinese - Best fit model.

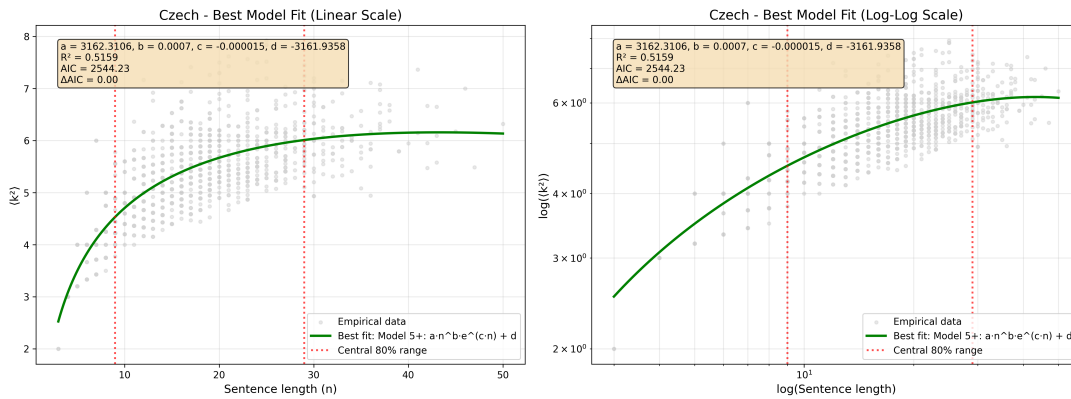


Figure 5: Czech - Best fit model.

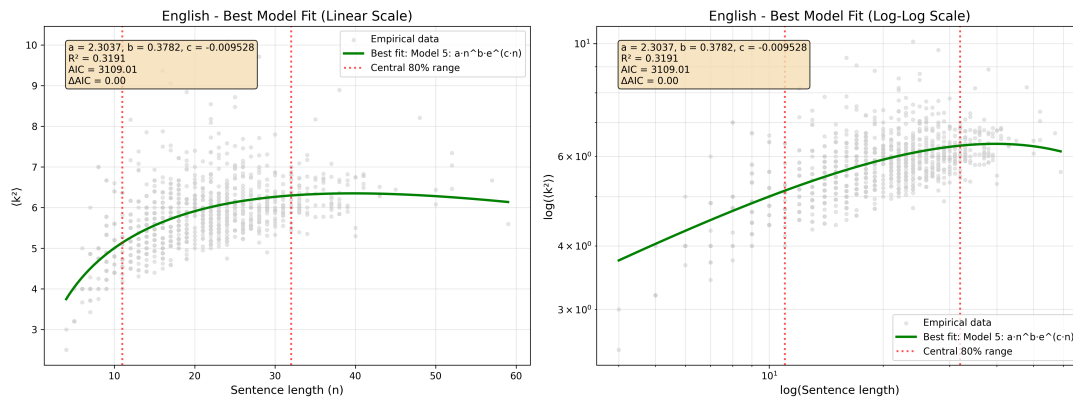


Figure 6: English - Best fit model.

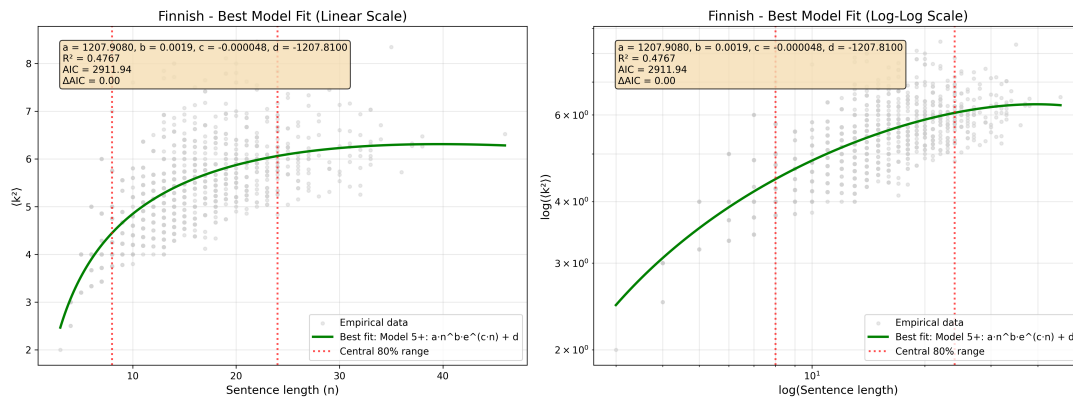


Figure 7: Finnish - Best fit model.

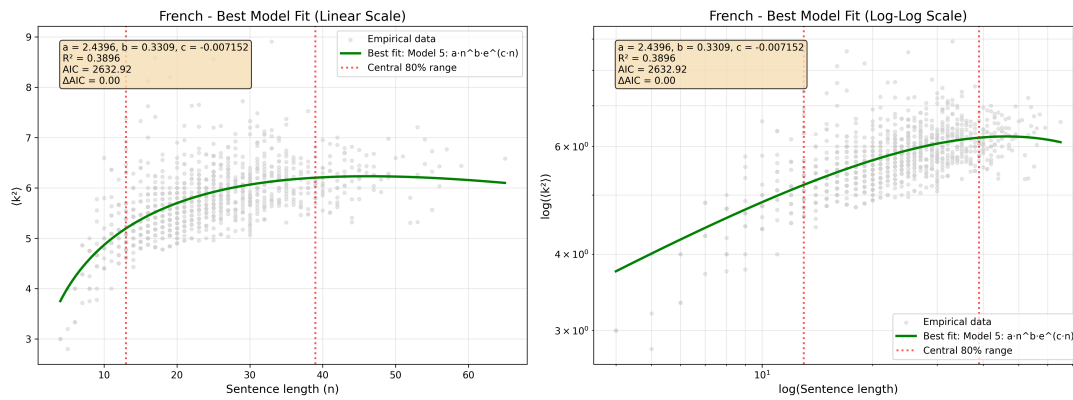


Figure 8: French - Best fit model.

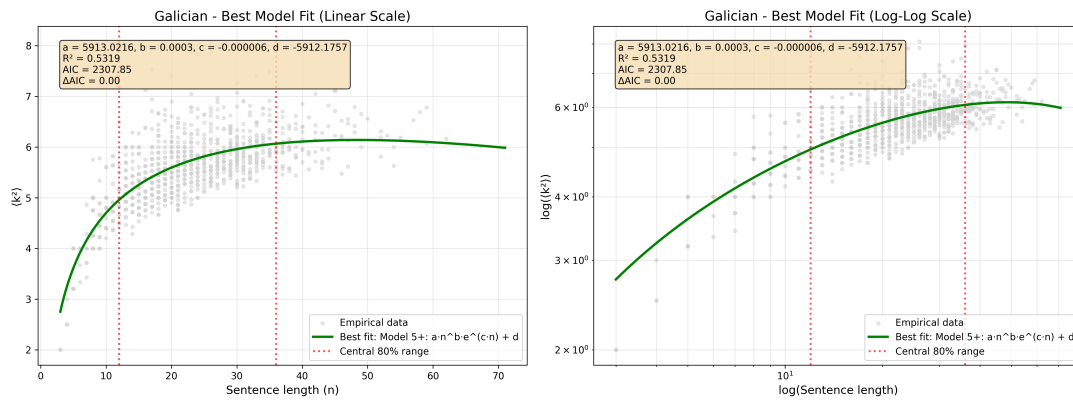


Figure 9: Galician - Best fit model.

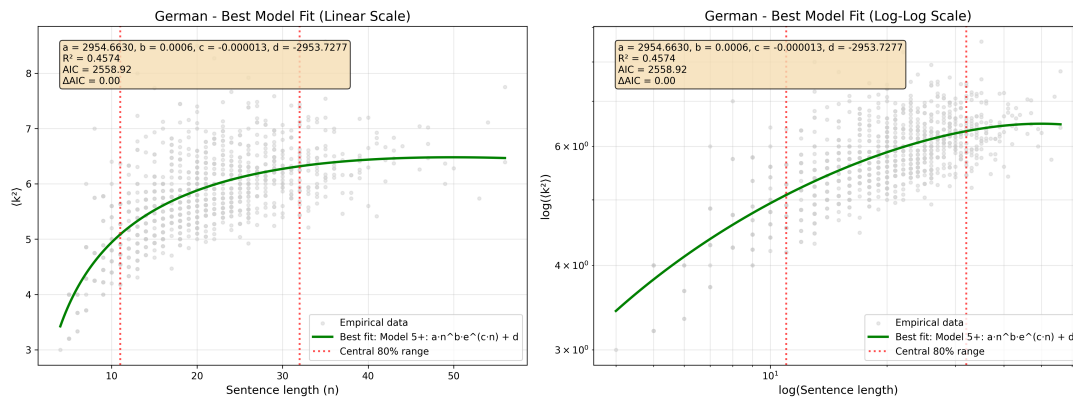


Figure 10: German - Best fit model.

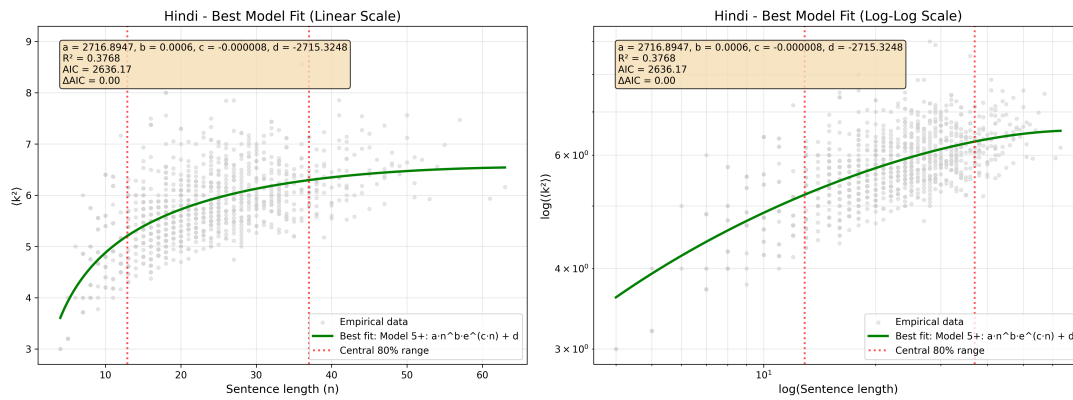


Figure 11: Hindi - Best fit model.

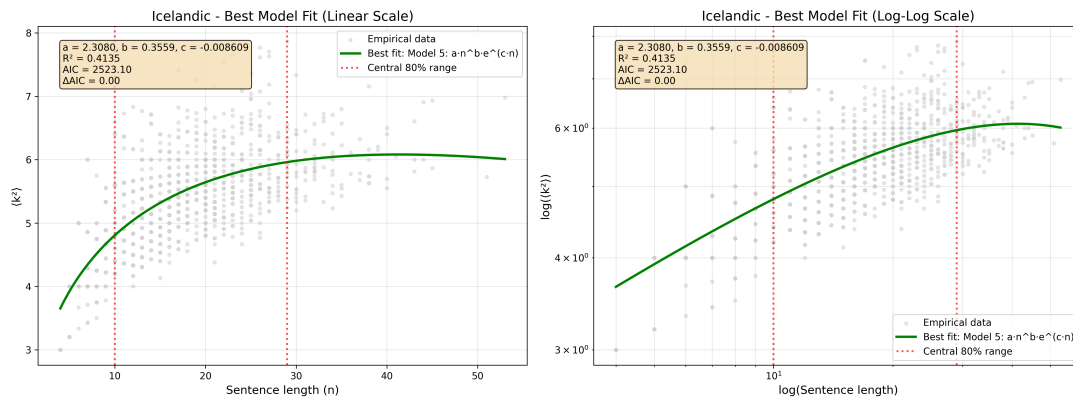


Figure 12: Icelandic - Best fit model.

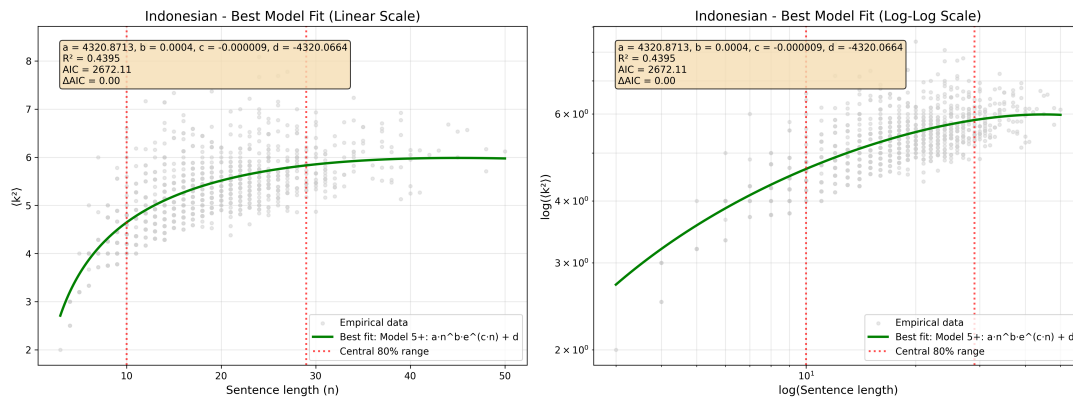


Figure 13: Indonesian - Best fit model.

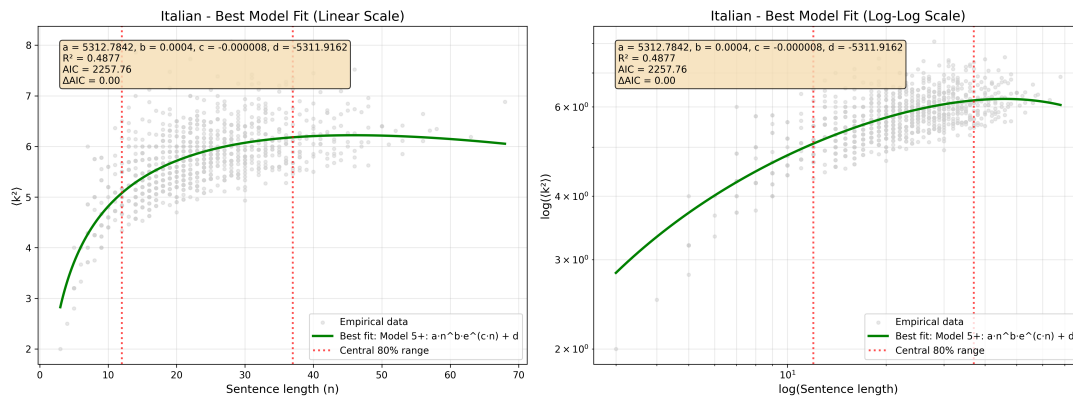


Figure 14: Italian - Best fit model.

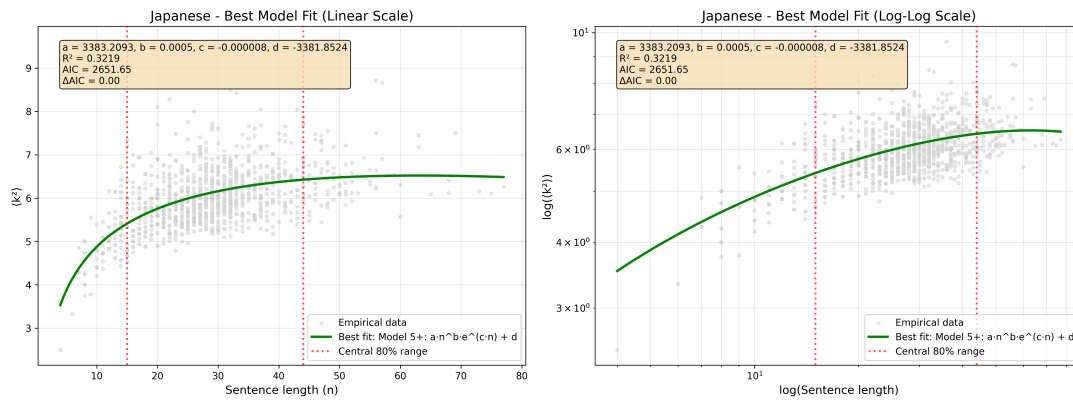


Figure 15: Japanese - Best fit model.

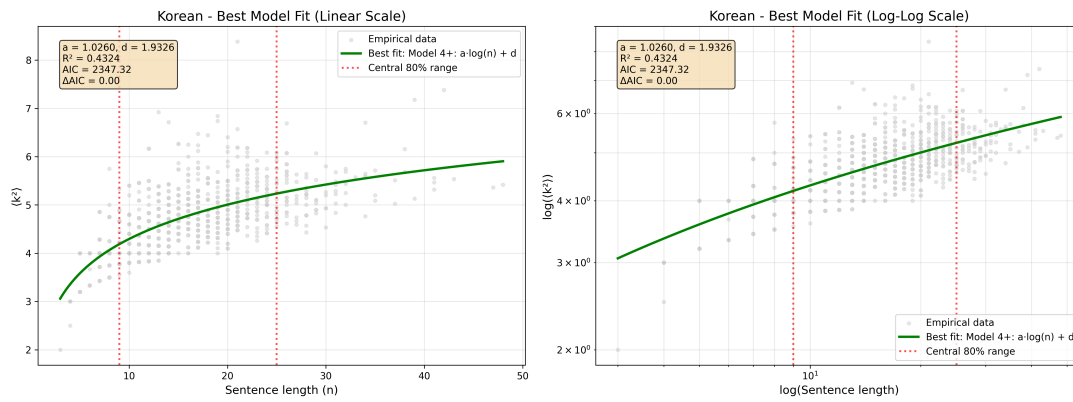


Figure 16: Korean - Best fit model.

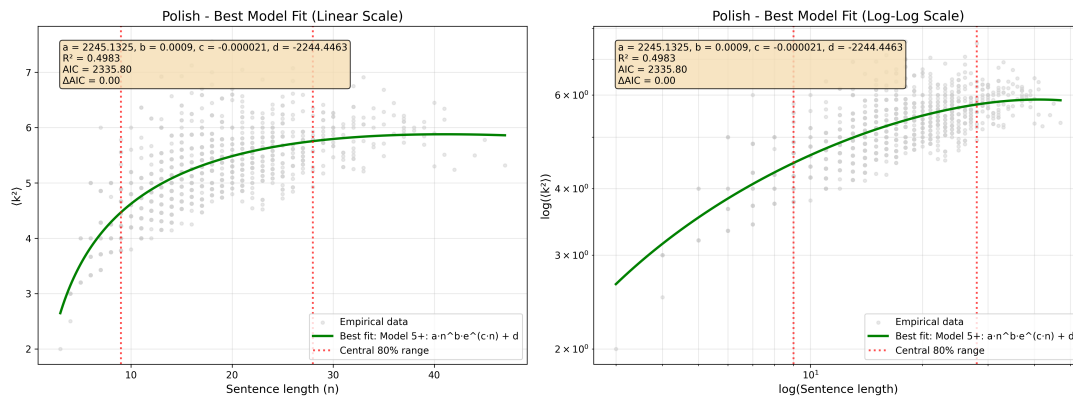


Figure 17: Polish - Best fit model.

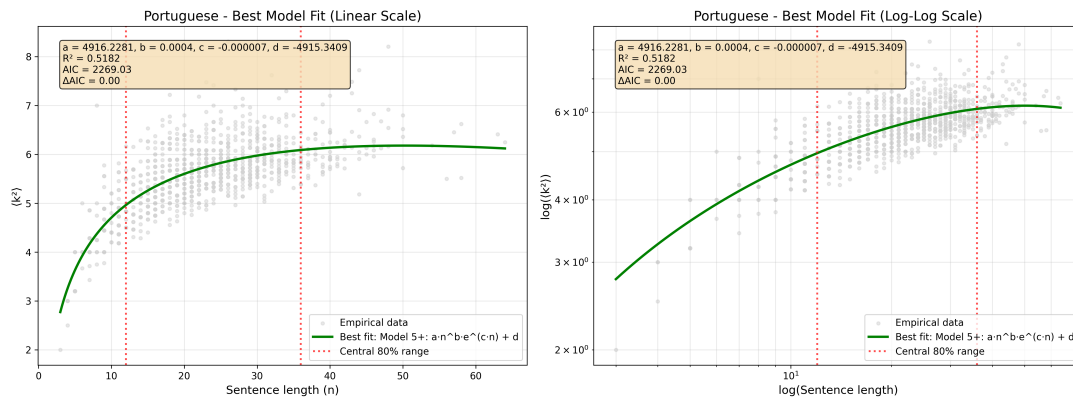


Figure 18: Portuguese - Best fit model.

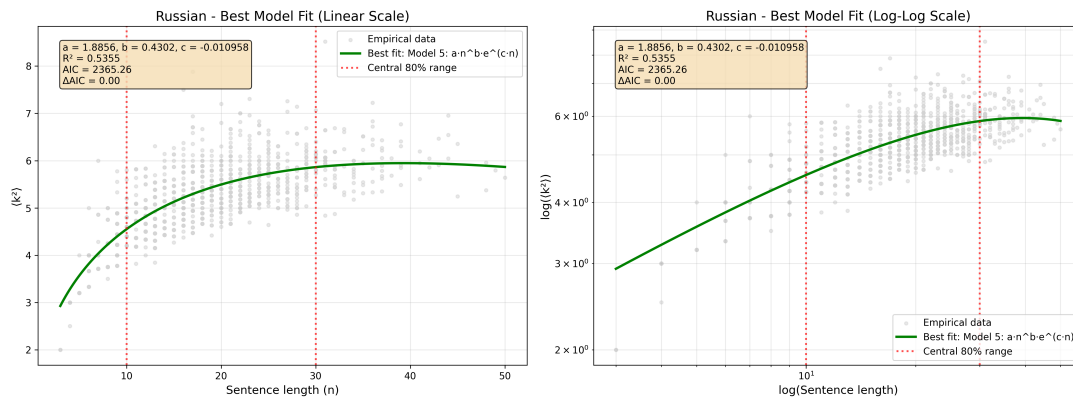


Figure 19: Russian - Best fit model.

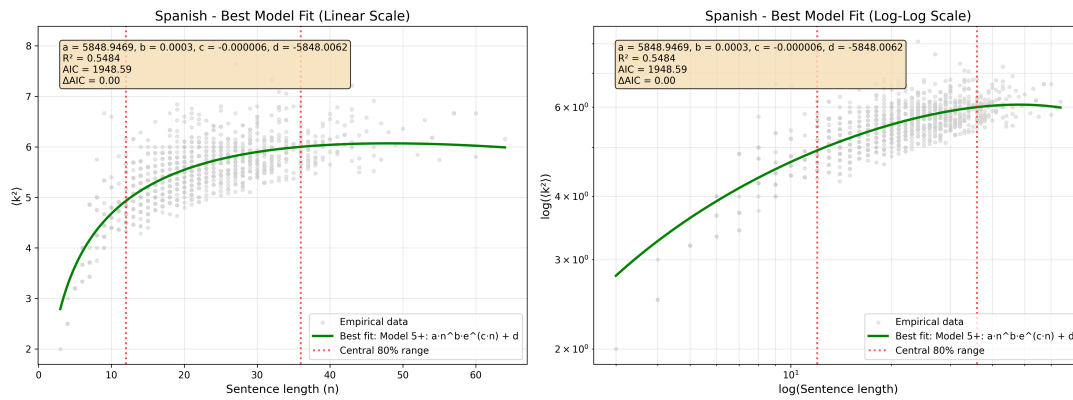


Figure 20: Spanish - Best fit model.

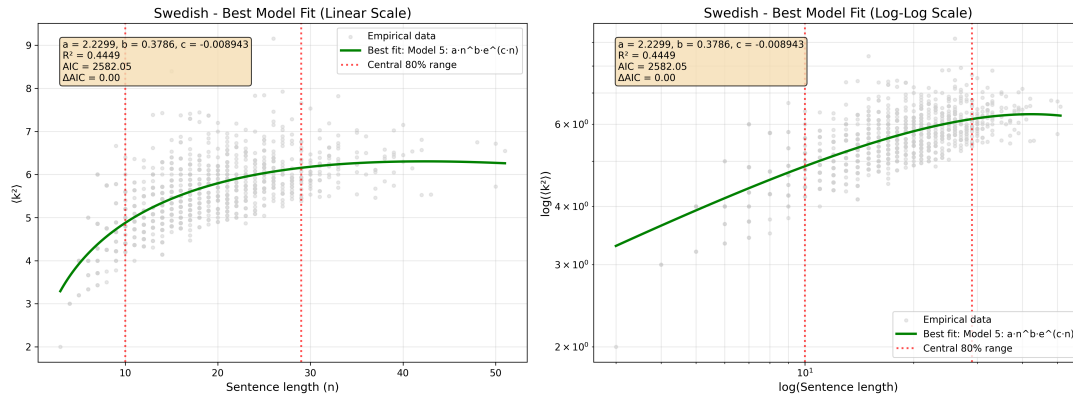


Figure 21: Swedish - Best fit model.

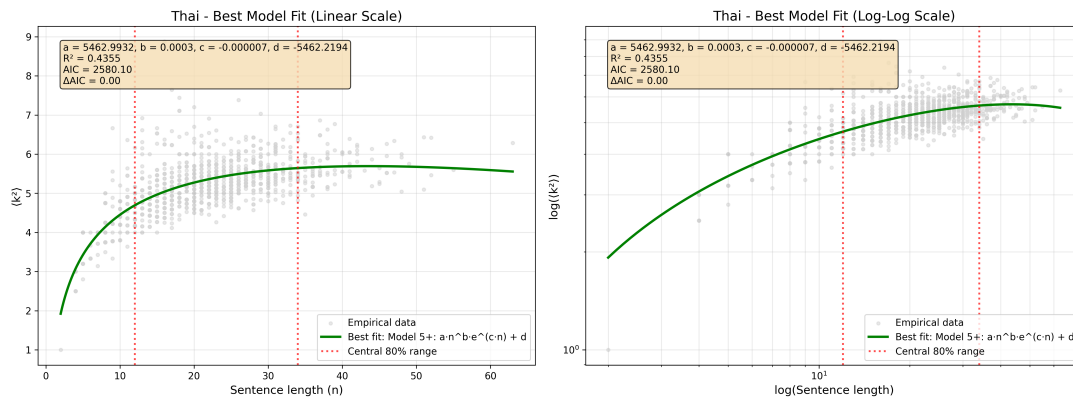


Figure 22: Thai - Best fit model.

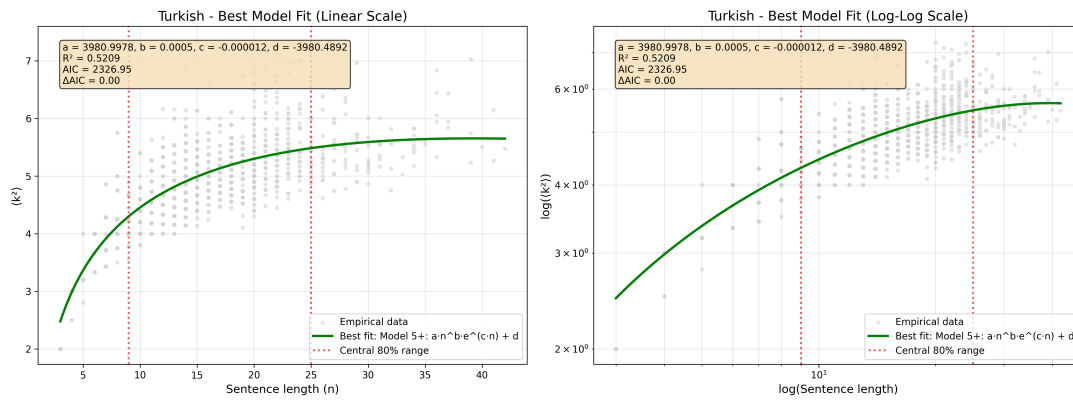


Figure 23: Turkish - Best fit model.

C Preliminary Analysis Plots

Preliminary analysis results for all 21 languages illustrate how $\langle k^2 \rangle$ scales with sentence length (n). The bottom-left panel shows the scatter plot together with the theoretical bounds in blue and the random-tree expectation in red. The top-right panel presents a log-log plot that reveals a sublinear scaling pattern, while the bottom-right panel displays averaged data that makes the underlying trends clearer.

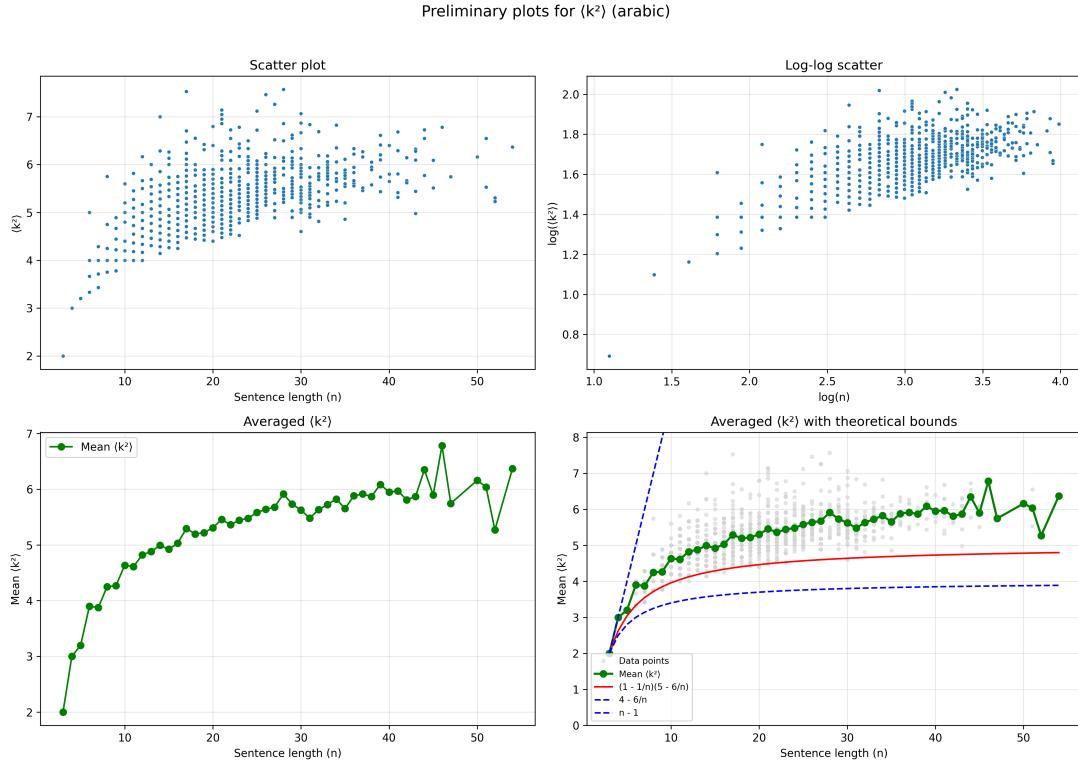


Figure 24: Preliminary results for Arabic.

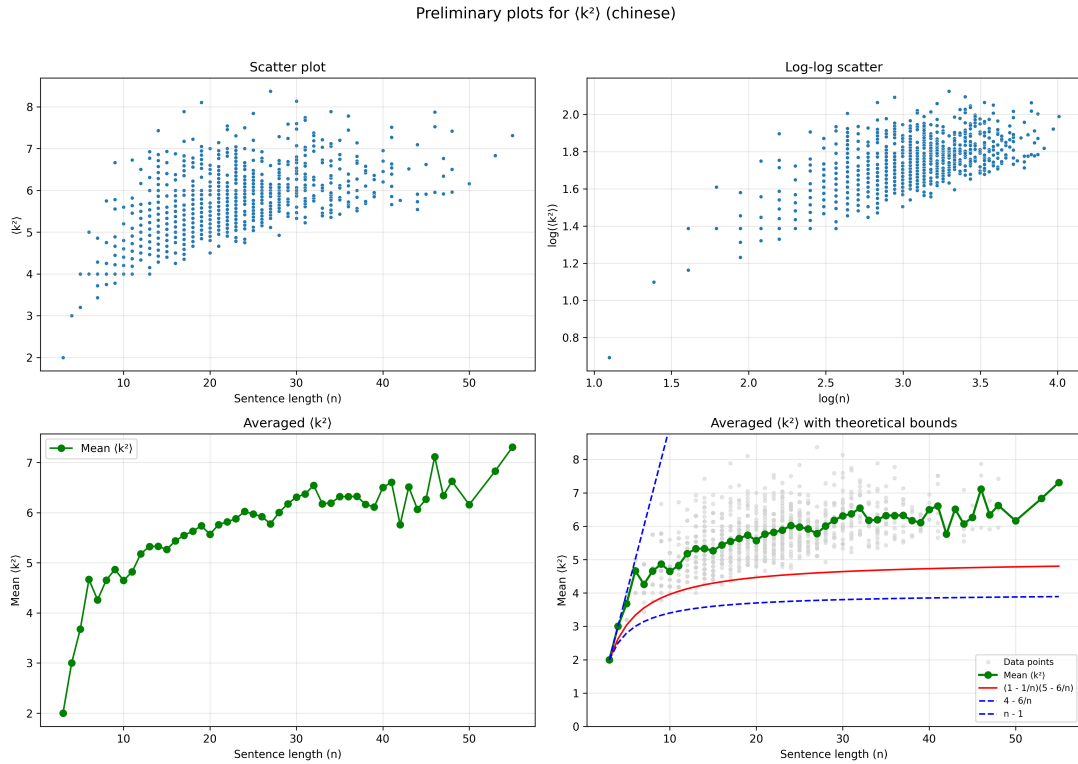


Figure 25: Preliminary results for Chinese.

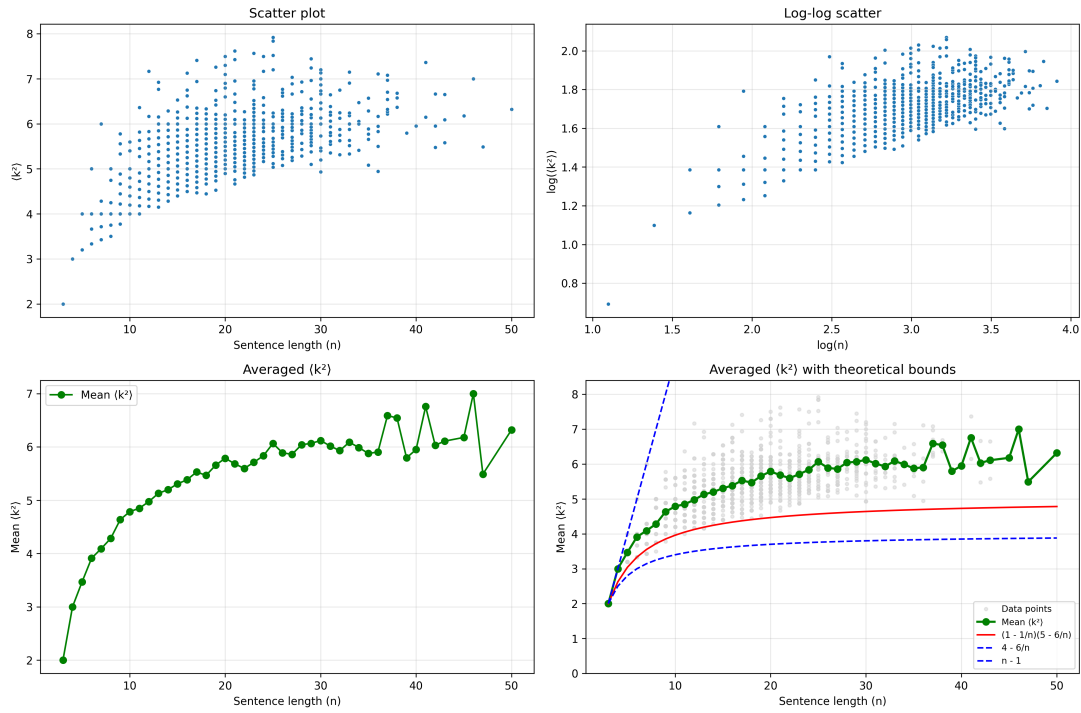
Preliminary plots for $\langle k^2 \rangle$ (czech)

Figure 26: Preliminary results for Czech.

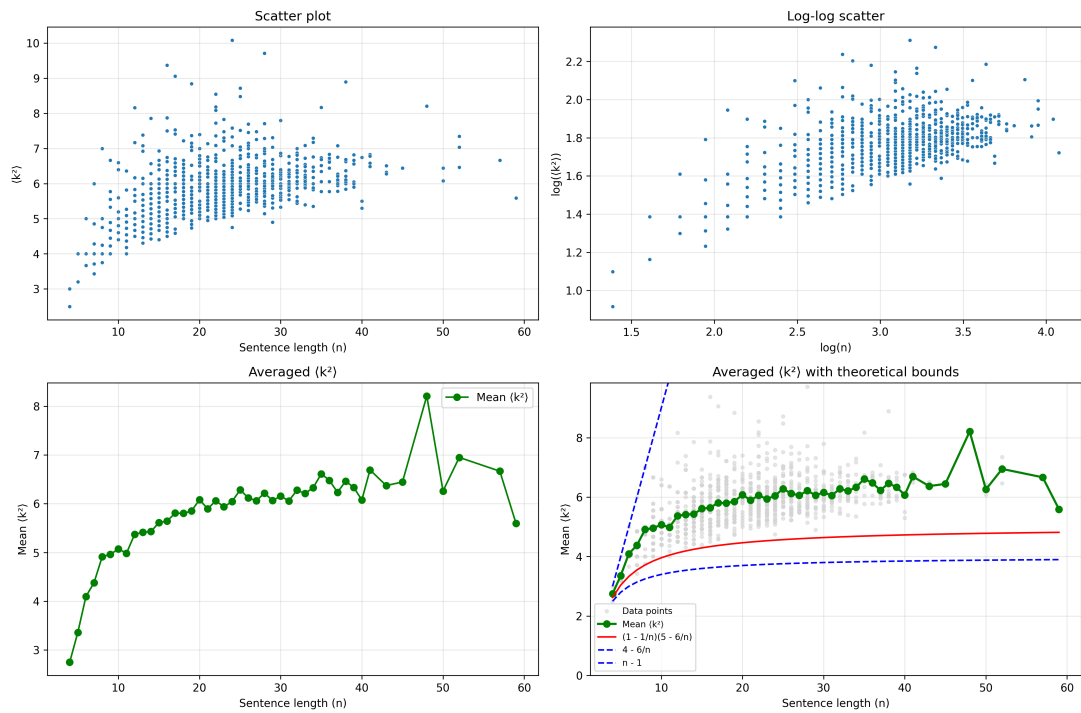
Preliminary plots for $\langle k^2 \rangle$ (english)

Figure 27: Preliminary results for English.

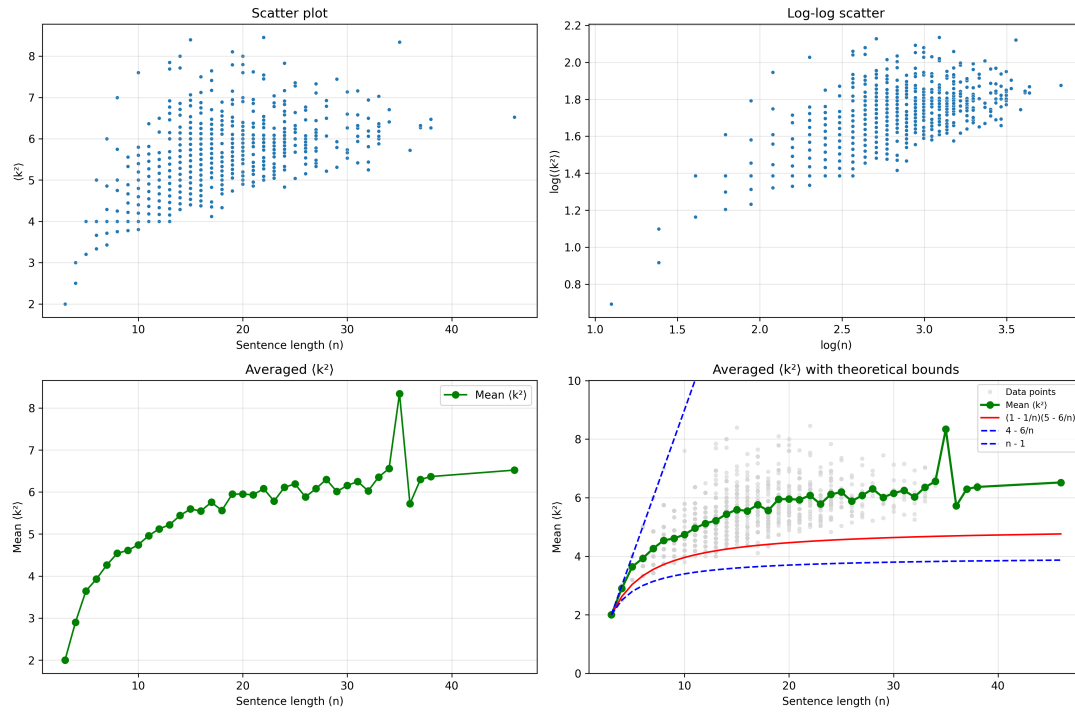
Preliminary plots for $\langle k^2 \rangle$ (finnish)

Figure 28: Preliminary results for Finnish.

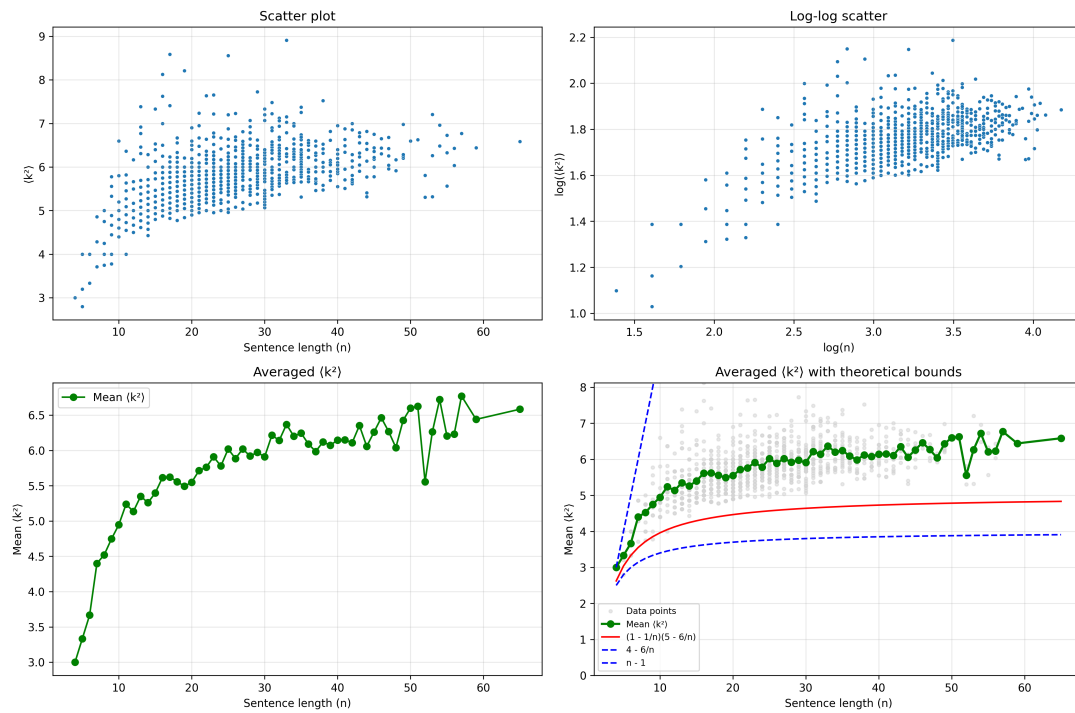
Preliminary plots for $\langle k^2 \rangle$ (french)

Figure 29: Preliminary results for French.

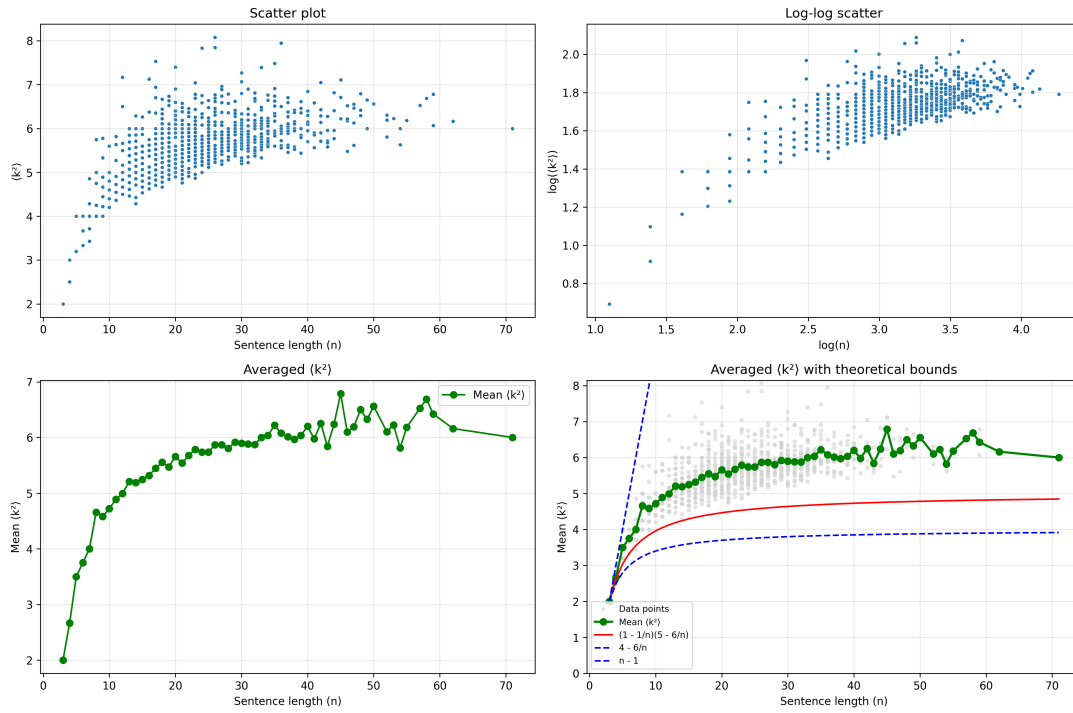
Preliminary plots for $\langle k^2 \rangle$ (galician)

Figure 30: Preliminary results for Galician.

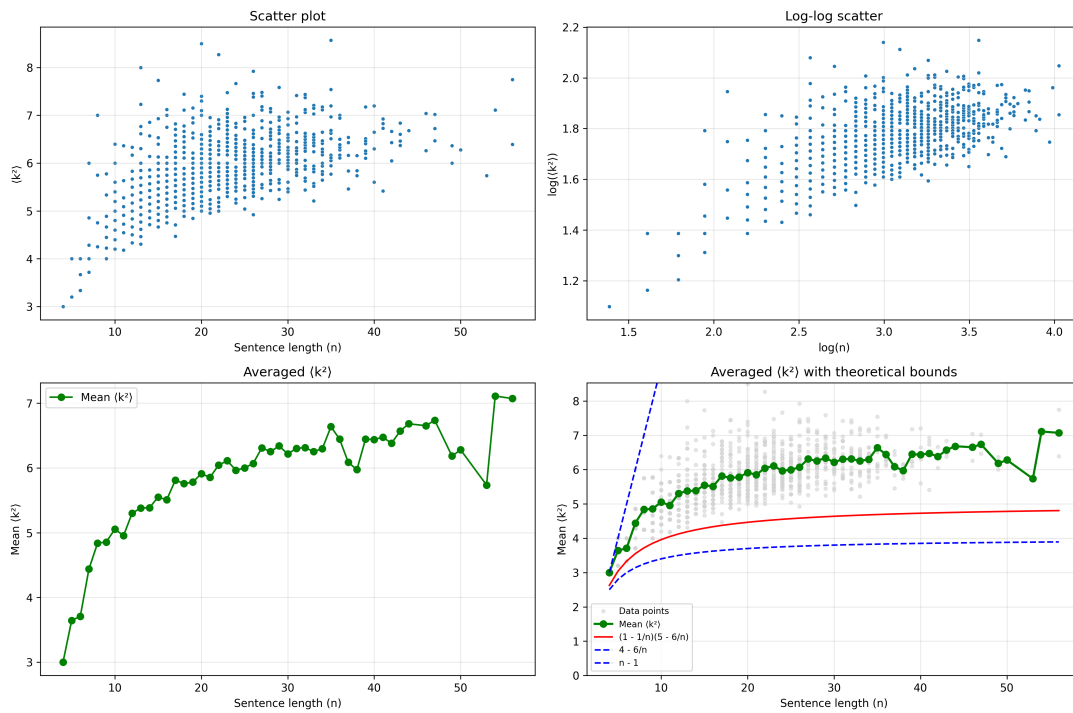
Preliminary plots for $\langle k^2 \rangle$ (german)

Figure 31: Preliminary results for German.

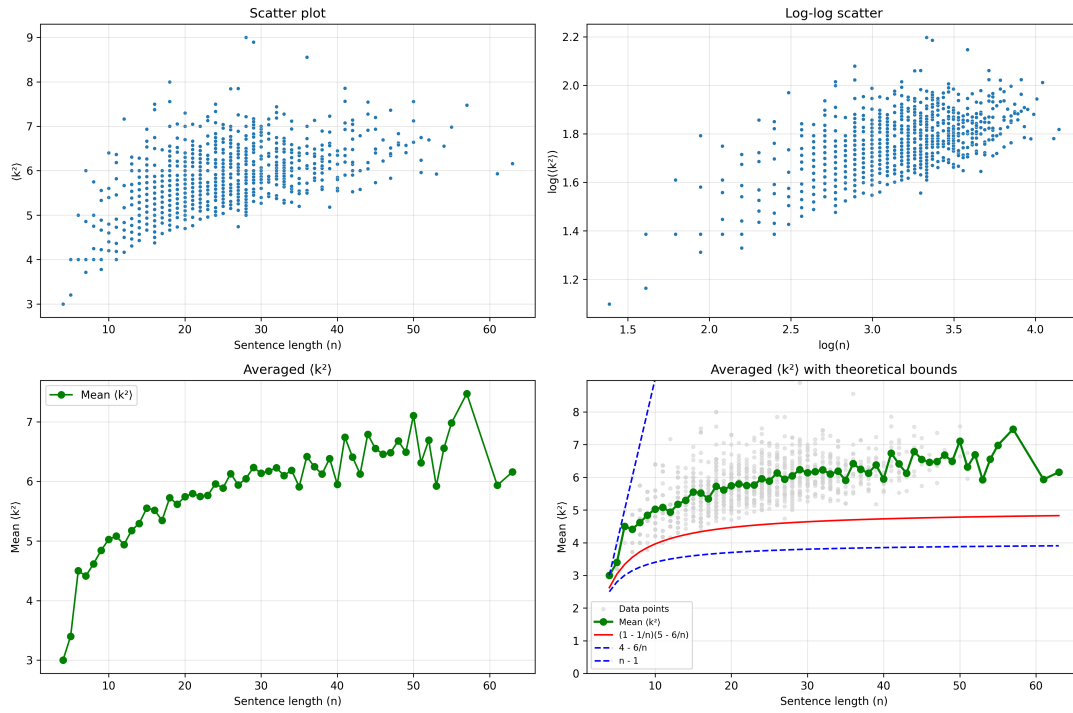
Preliminary plots for $\langle k^2 \rangle$ (hindi)

Figure 32: Preliminary results for Hindi.

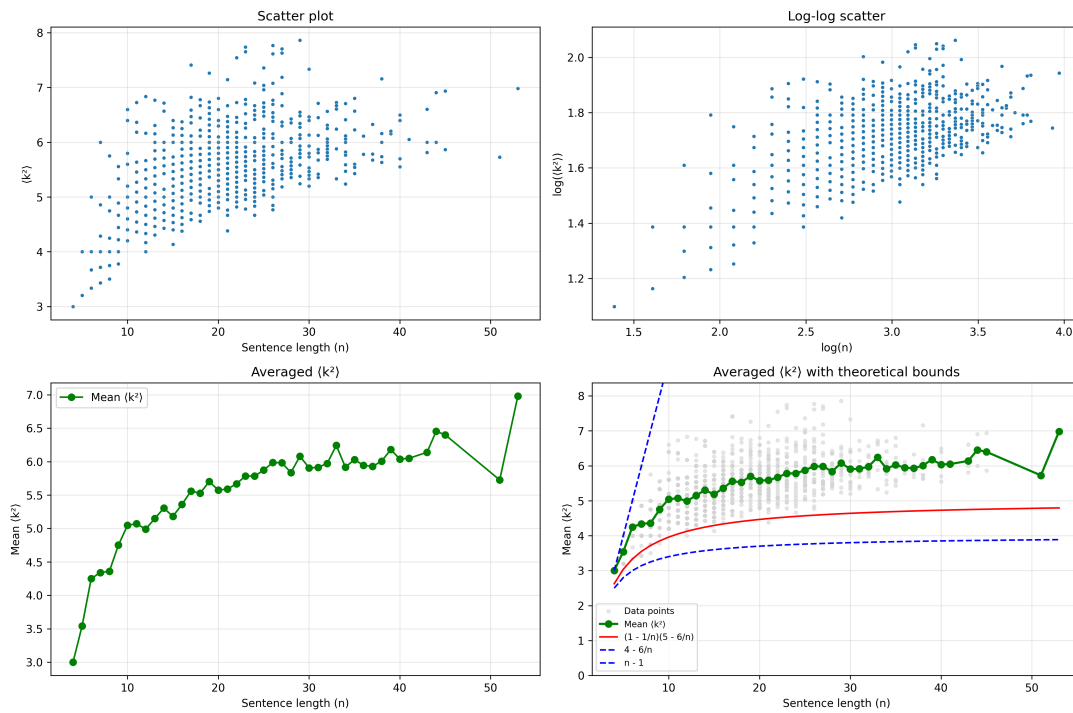
Preliminary plots for $\langle k^2 \rangle$ (icelandic)

Figure 33: Preliminary results for Icelandic.

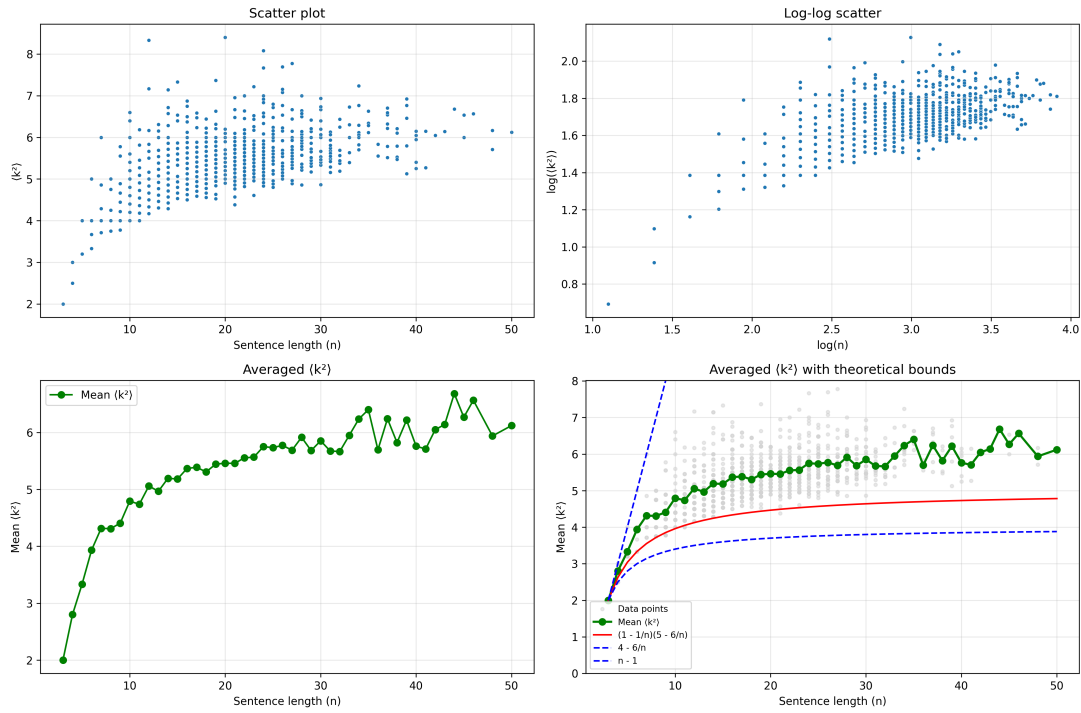
Preliminary plots for $\langle k^2 \rangle$ (indonesian)

Figure 34: Preliminary results for Indonesian.

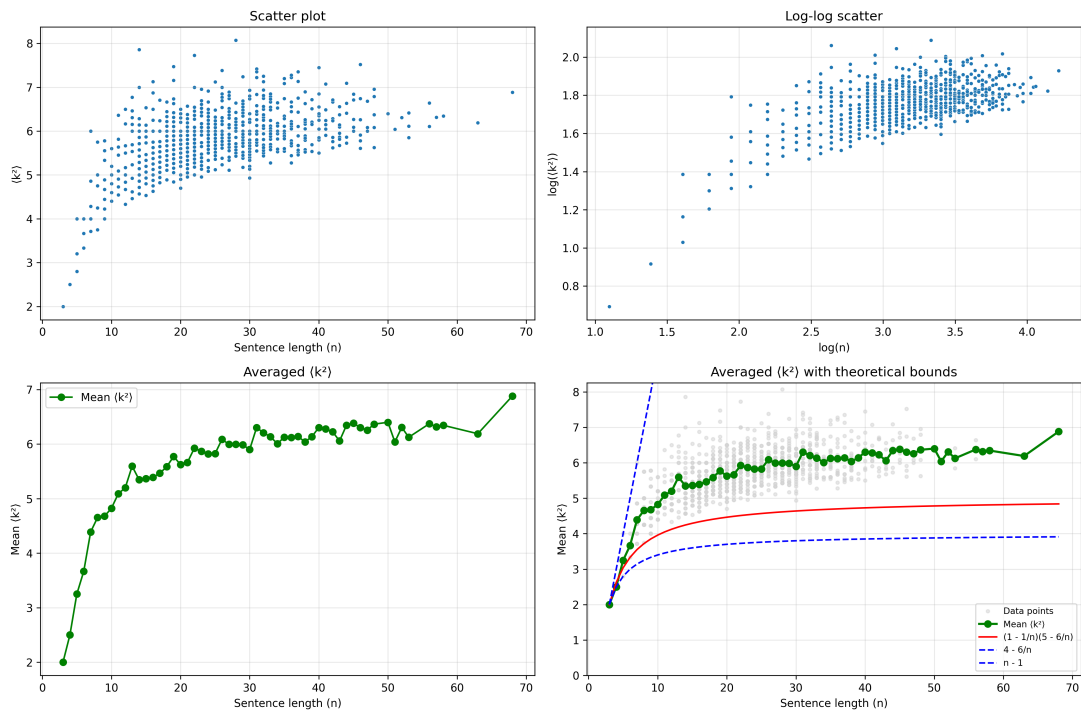
Preliminary plots for $\langle k^2 \rangle$ (italian)

Figure 35: Preliminary results for Italian.

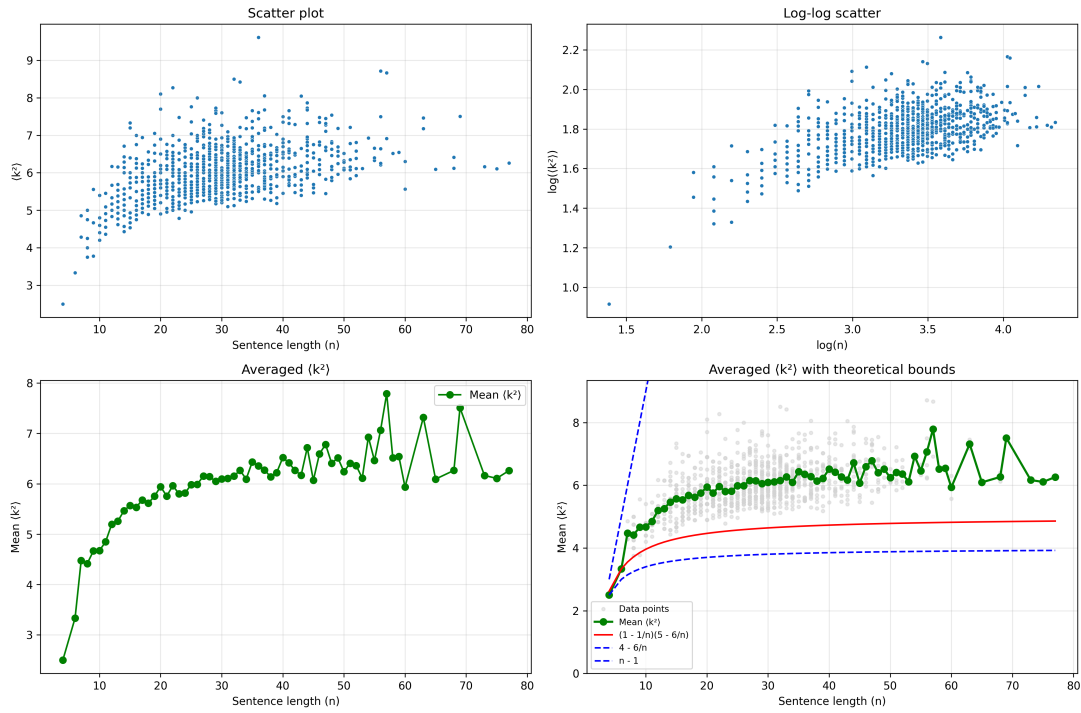
Preliminary plots for $\langle k^2 \rangle$ (japanese)

Figure 36: Preliminary results for Japanese.

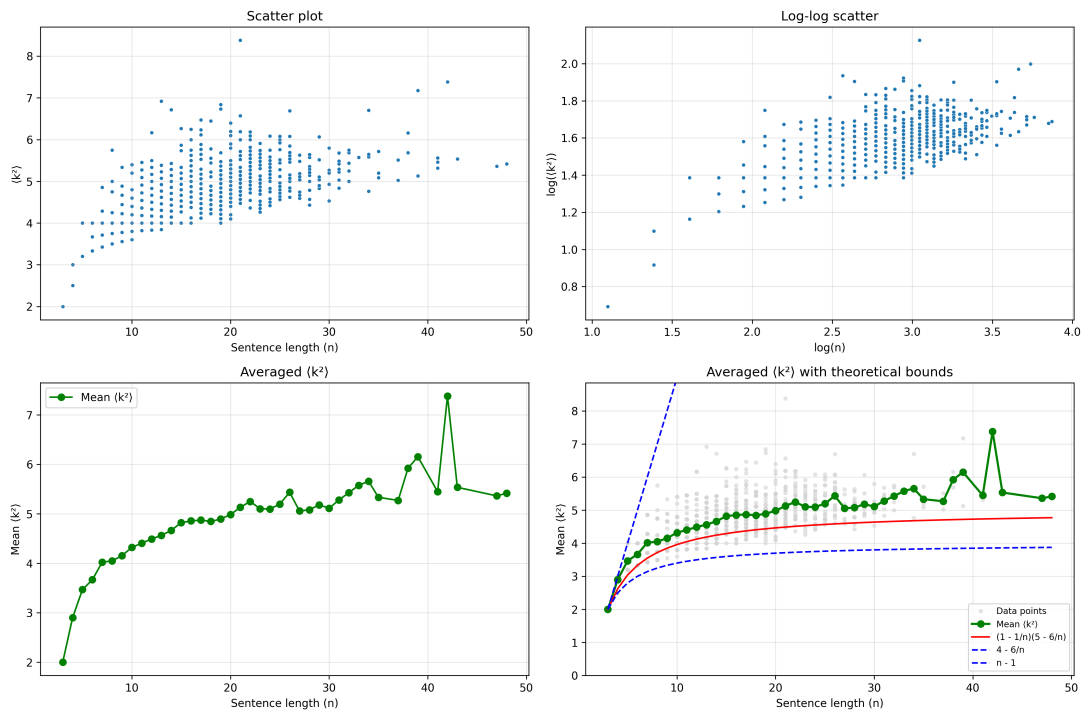
Preliminary plots for $\langle k^2 \rangle$ (korean)

Figure 37: Preliminary results for Korean.

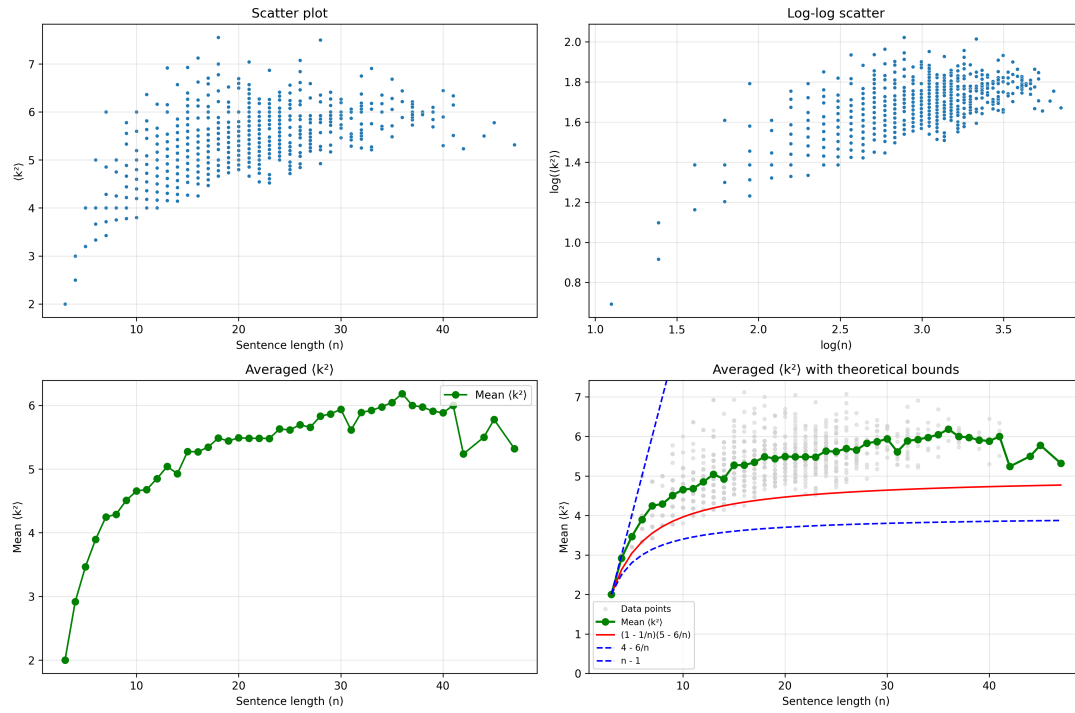
Preliminary plots for $\langle k^2 \rangle$ (polish)

Figure 38: Preliminary results for Polish.

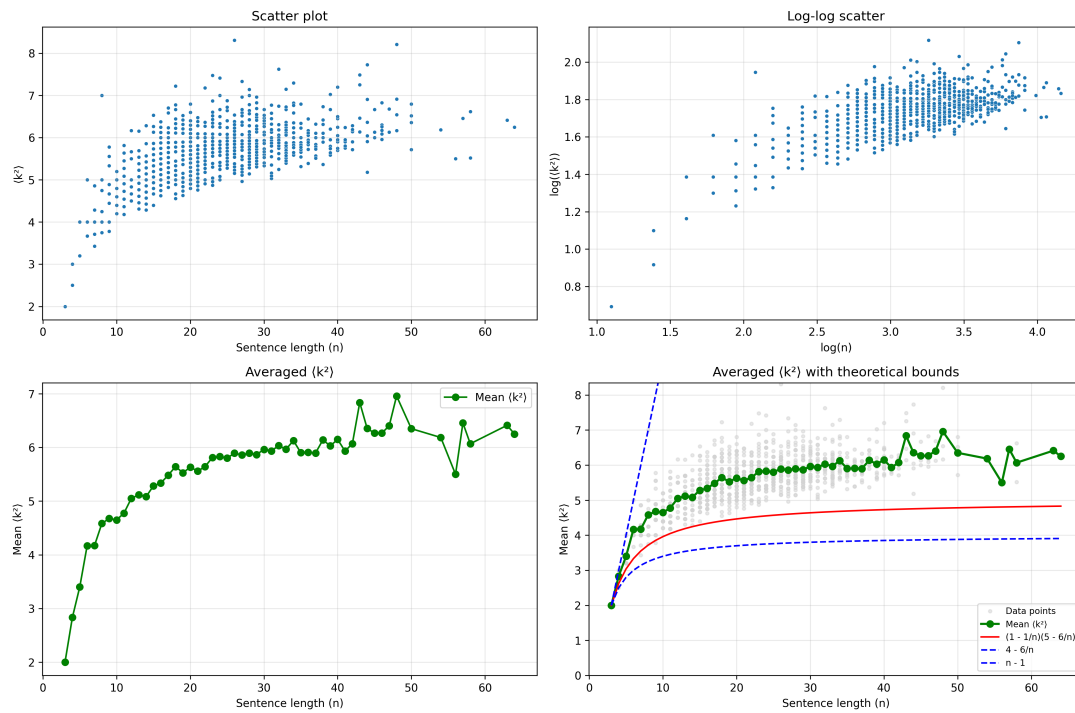
Preliminary plots for $\langle k^2 \rangle$ (portuguese)

Figure 39: Preliminary results for Portuguese.

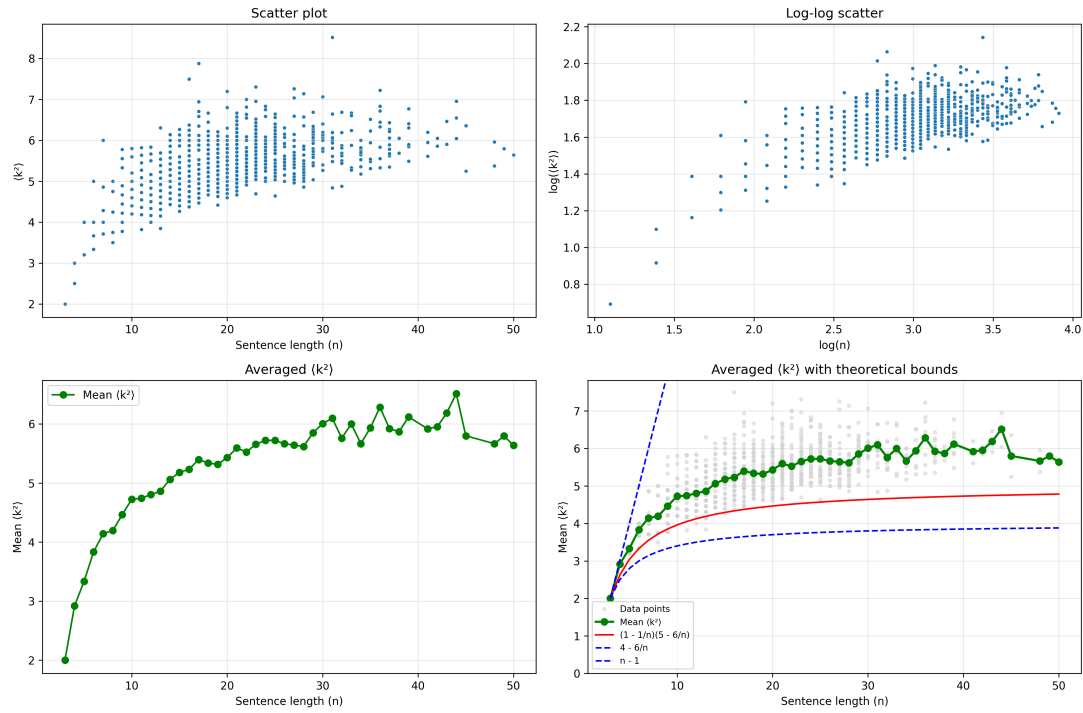
Preliminary plots for $\langle k^2 \rangle$ (russian)

Figure 40: Preliminary results for Russian.

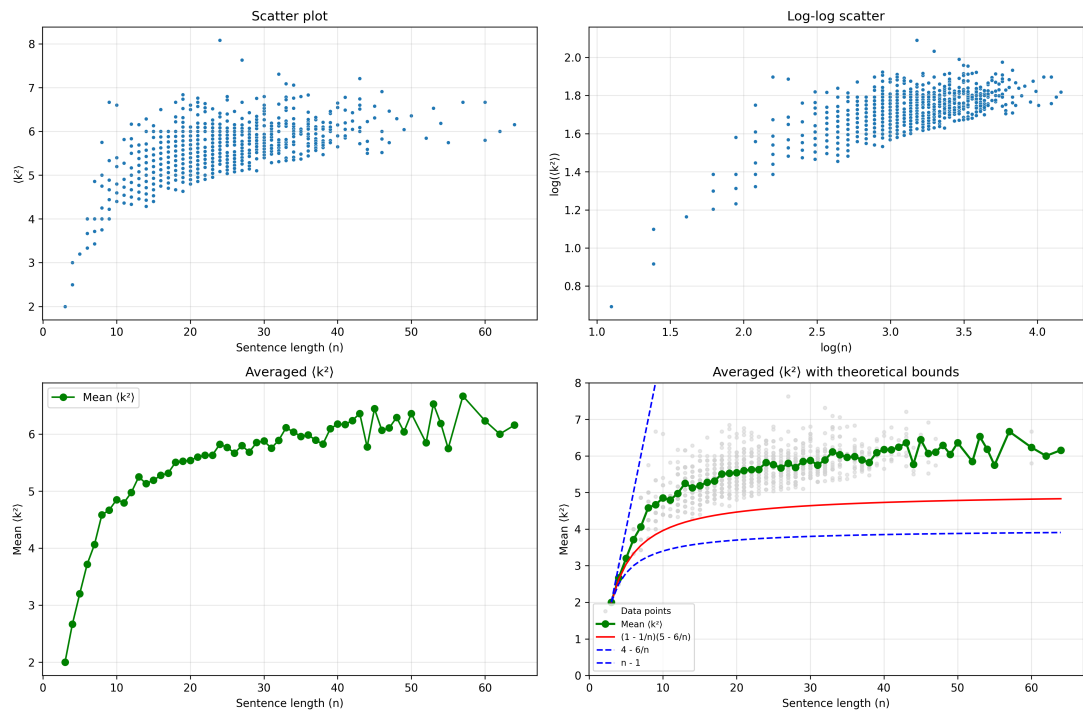
Preliminary plots for $\langle k^2 \rangle$ (spanish)

Figure 41: Preliminary results for Spanish.

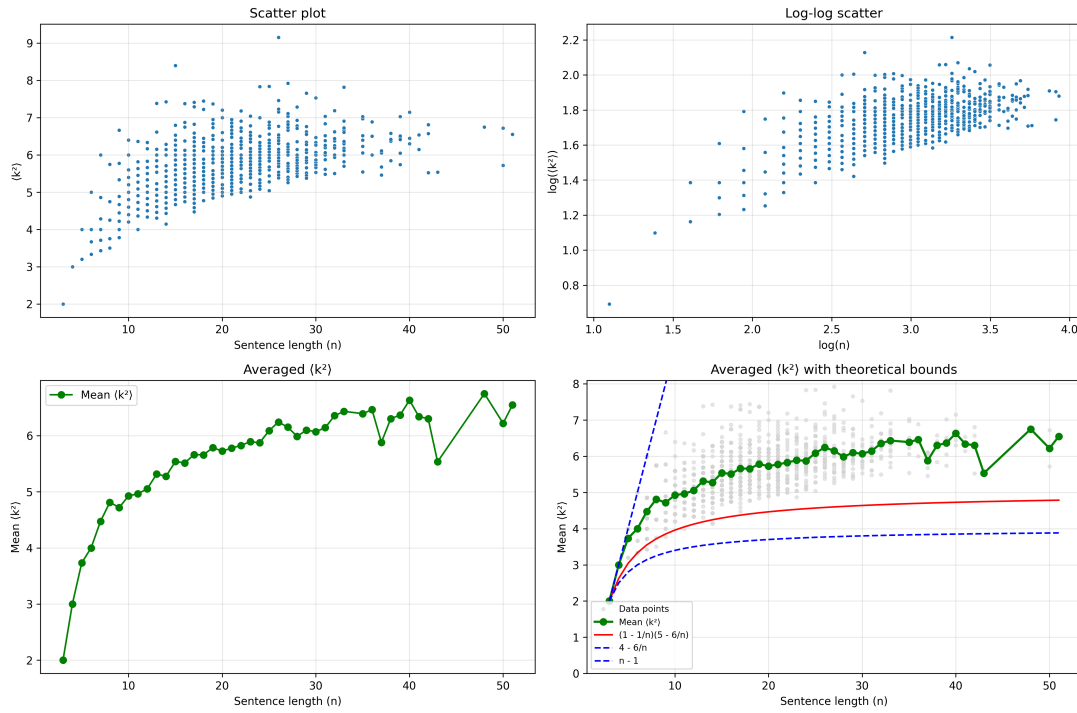
Preliminary plots for $\langle k^2 \rangle$ (swedish)

Figure 42: Preliminary results for Swedish.

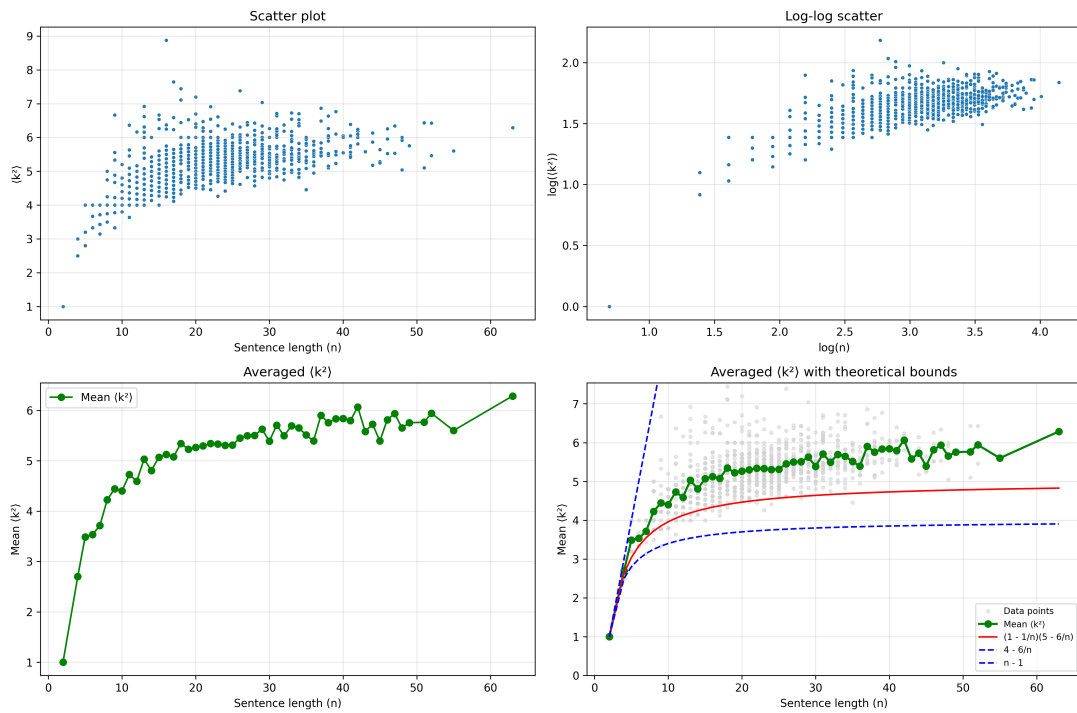
Preliminary plots for $\langle k^2 \rangle$ (thai)

Figure 43: Preliminary results for Thai.

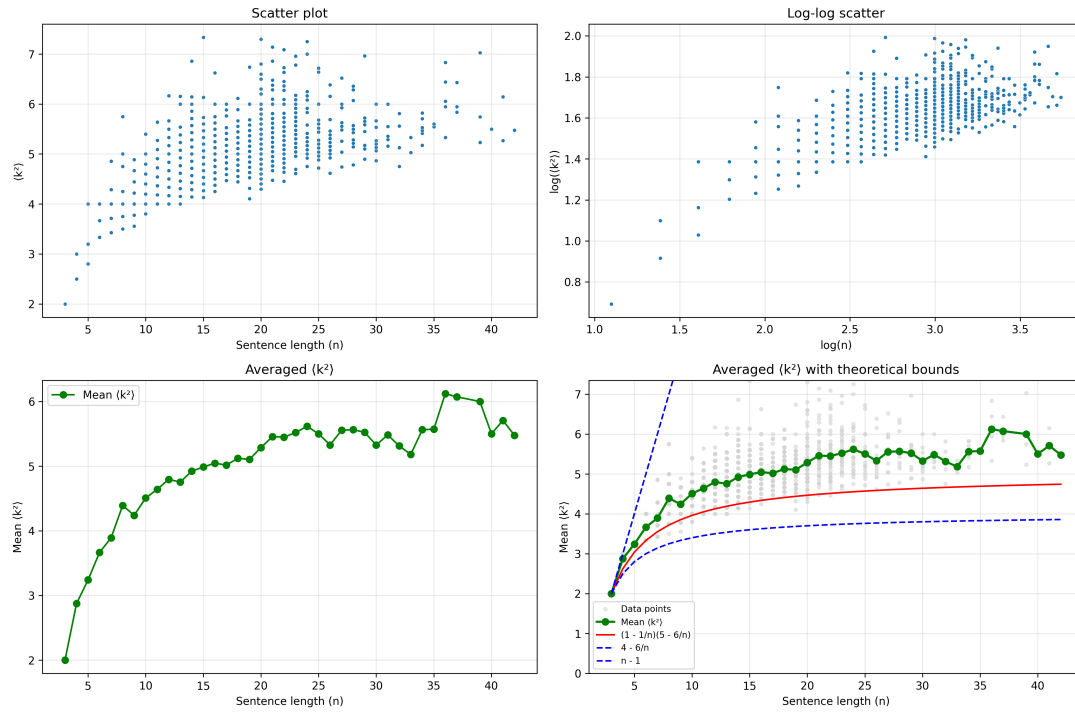
Preliminary plots for $\langle k^2 \rangle$ (turkish)

Figure 44: Preliminary results for Turkish.